MÉTODO DE LOS DESPLAZAMIENTOS

1.- $\theta_E = 2 \times 10^{-3}$ rad

$\mu_E = -4 \times 10^{-2}$m

$EKr = 10^5$ Kg.m

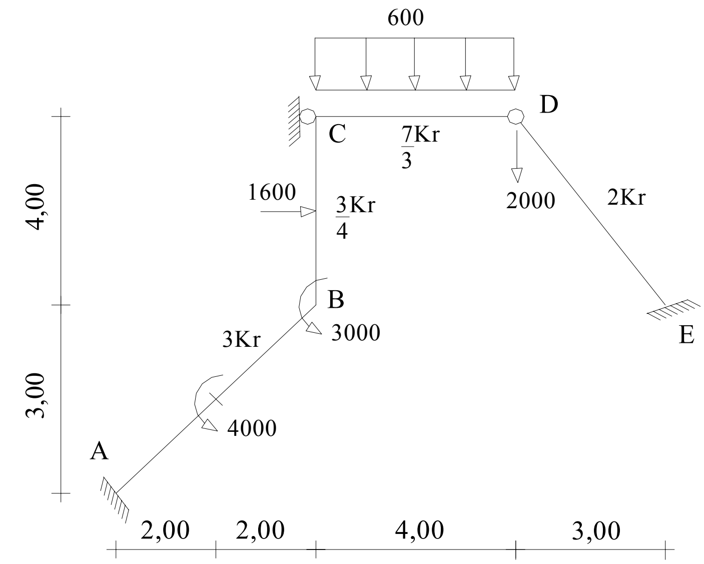
$\Delta t_o^{AB} = 10°C$

$\alpha t = 1 \times 10^{-5}\ °C^{-1}$

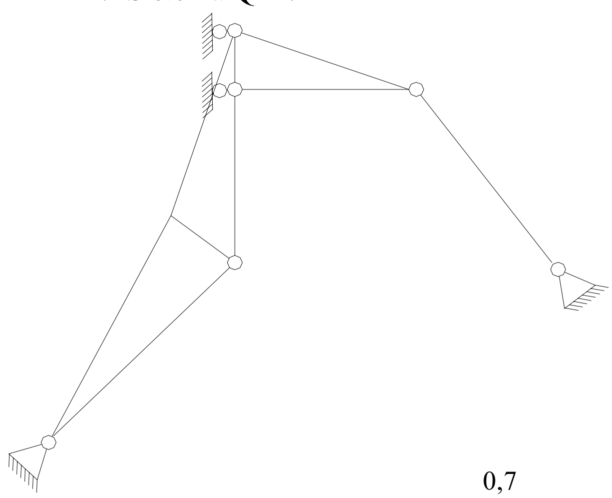
Resolver:

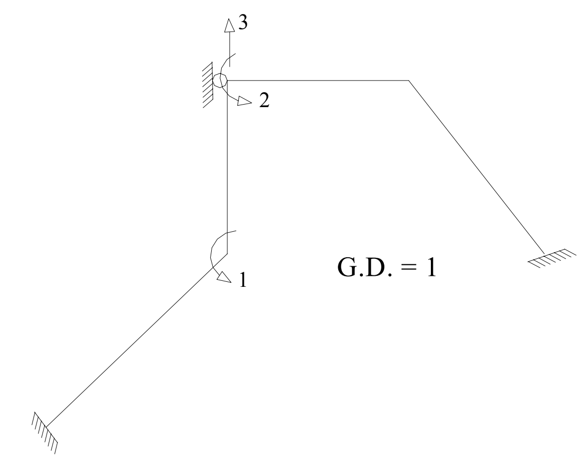
a.- Método de Cross.

b.- Método de las Rotaciones.

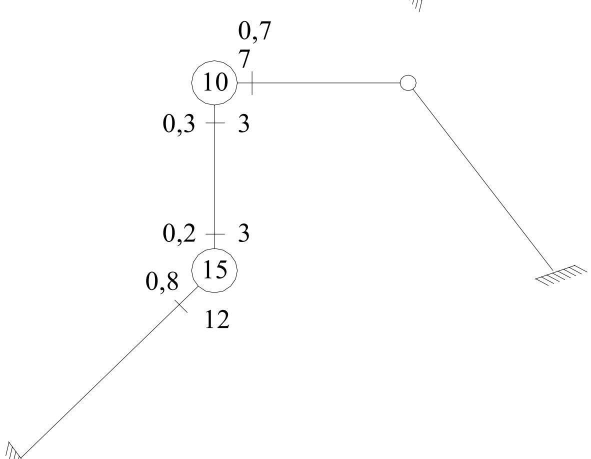


1.- Sistema Q-D.





2.- D_{ij} y r_{ij}



3.- Problemas a Resolver (2).

4.- Problema # 1.

4.1.-Problema Complementario: $\theta_B = \theta_C = 0$

4.1.1.- $M^E_{ij} = F$ (cargas, $\Delta(\Delta t)$); $\Delta(\Delta t) = 0$

$$M^E_{AB} = 1000; \qquad M^E_{BA} = 1000$$

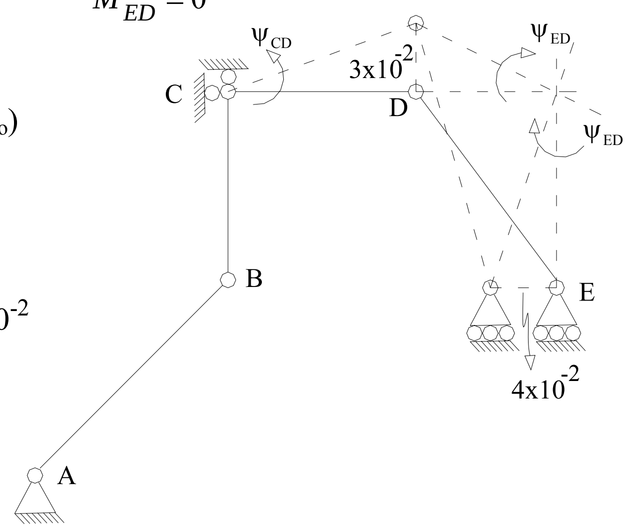
$$M^E_{BC} = \frac{1600x4}{8} = 800; \qquad M^E_{CB} = \frac{-1600x4}{8} = -800$$

$$M^E_{CD} = \frac{600x4^2}{12} = 800; \qquad M^E_{DC} = \frac{-600x4^2}{12} = -800$$

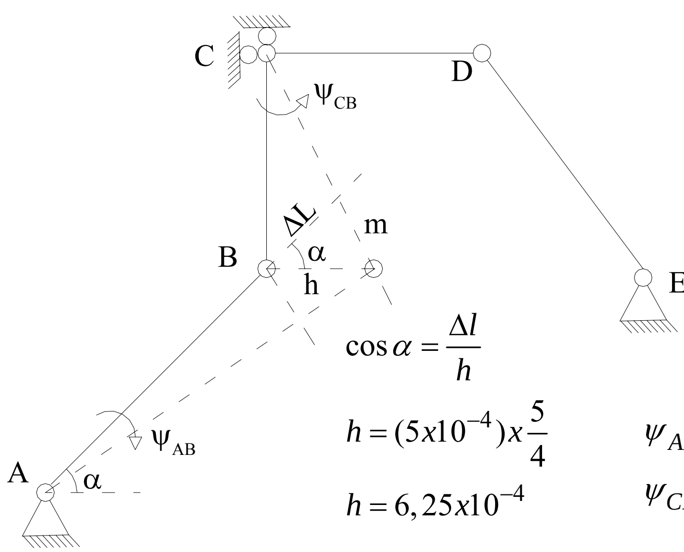
$$M^E_{DE} = 0; \qquad M^E_{ED} = 0$$

4.1.2.- $\psi^0_{ij} =$ F (mov. Apoyo, Δt_o)

a) F (mov. apoyo)



$$\left\{ \psi_{CD} = \frac{3}{4} \text{ x } 10^{-2}; \right. \qquad \left\{ \psi_{ED} = -1 \text{ x } 10^{-2} \right.$$



b) F (Δt_o) $\rightarrow$ (A-B)

$$tg\alpha = \frac{m}{\Delta l} \rightarrow m = \Delta l . tg\alpha$$

$$m = (5x10^{-4})x\frac{3}{4}$$

$$m = 3,75x10^{-4}$$

$$\cos\alpha = \frac{\Delta l}{h}$$

$$h = (5x10^{-4})x\frac{5}{4}$$

$$h = 6,25x10^{-4}$$

$$\psi_{AB} = -7,5 \text{ x } 10^{-5}$$

$$\psi_{CB} = 1,5625 \text{ x } 10^{-4}$$

$$\begin{cases} \psi^o_{AB} = -7,5 \text{ x } 10^{-5} \\ \psi^o_{CB} = 1,5625 \text{ x } 10^{-4} \end{cases} \qquad \begin{cases} \psi^o_{CD} = 7,5 \text{ x } 10^{-3} \\ \psi^o_{ED} = -1 \text{ x } 10^{-2} \end{cases}$$

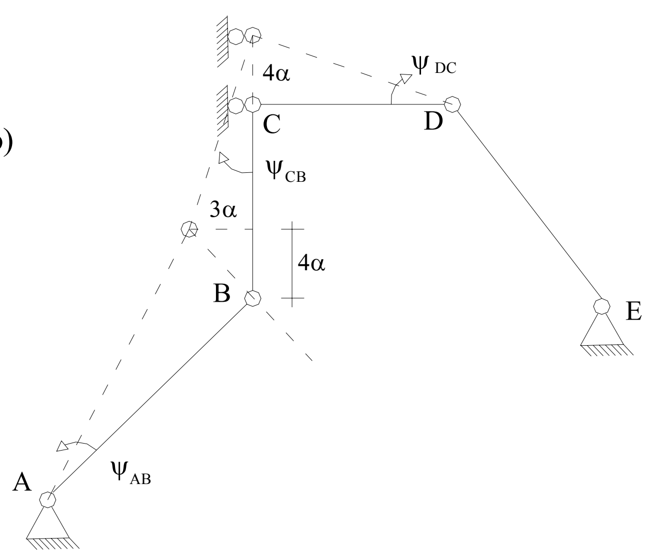
4.1.3.- $M^0_{ij} = ?$

$$\begin{cases} M^0_{AB} = 1000 - 18EKr(-7,5x10^{-5}) = 1135 \\ M^0_{BA} = 1000 - 18EKr(-7,5x10^{-5}) = 1135 \\ M^0_{BC} = 800 - 4,5EKr(1,5625x10^{-4}) = 730 \\ M^0_{CB} = -800 - 4,5EKr(1,5625x10^{-4}) = -870 \\ M'^0_{CD} = 800 + \frac{1}{2}(0+800) + 7EKr(0 - 7,5x10^{-3}) = -4050 \\ M'^0_{ED} = \frac{1}{2}(0-0) + 6EKr(2x10^{-3} + 1x10^{-2}) = 7200 \end{cases}$$

$$\begin{cases} M^{Fij}_B = 1135 + 730 - 3000 = -1135 \\ M^{Fij}_C = -870 - 4050 = -4920 \end{cases}$$

5.- Problema # 2.

5.1.- $\psi^0_{ij} =$ F (Trasl. Apoyo)



$$\begin{cases} \psi_{AB} = \alpha \\ \psi_{CD} = -\alpha \\ \psi_{CB} = (-3/4)\alpha \end{cases}$$

5.2.- $M^o_{ij} =$ F (ψ_{ij})

$$\begin{cases} M'^0_{ED} = 0 \\ M'^0_{CD} = 3E(\frac{7}{3}Kr)(+\alpha) = 7EKr\alpha \\ M^0_{CB} = M^0_{BC} = -6E(\frac{3}{4}Kr)(\frac{-3}{4}\alpha) = (3,375)EKr\alpha \\ M^0_{AB} = M^0_{BA} = -6E(3Kr)\alpha = -18EKr\alpha \end{cases}$$

Sea EKr α = 1000x

$$\begin{cases} M'^0_{CD}/x = 7000 \\ M^0_{AB}/x = M^0_{BA}/x = -18000 \end{cases} \qquad \begin{cases} M^0_{CB}/x = M^0_{BC}/x = 3375 \\ M'^0_{ED}/x = 0 \end{cases}$$

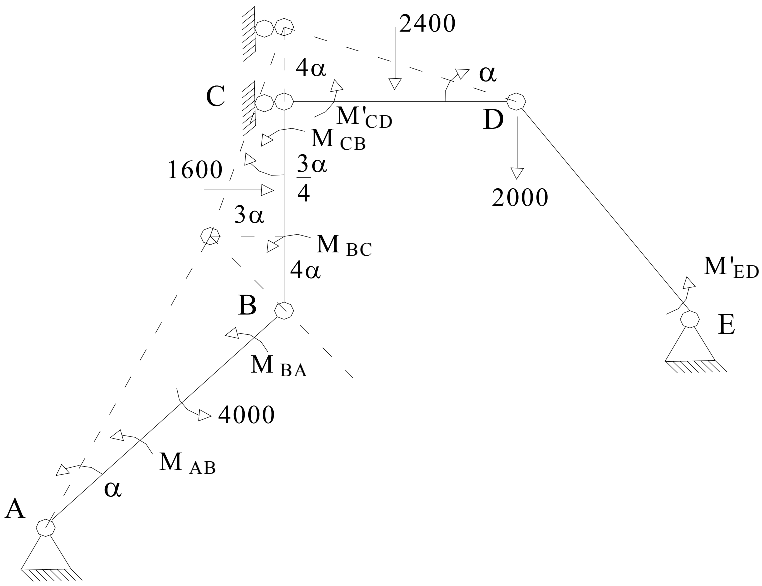
5.3.- $\left\{ M_B^{Fij} / x = -14625; \qquad M_C^{Fij} / x = 10375 \right.$

6.- Solución de las Etapas Complementarias

Junta	A	B		C	
Miembro	AB	BA	BC	CB	CD
D_{ij}	0	0,8	0,2	0,3	0,7
M_{ij}^{0} (1)	1135	1135	730	-870	-4050
			738	1476	3444
M_{ij}^{C} (1)	159	318	79	40	
			-6	-12	-28
	3	5	1		
M_{ij} (1)	1297	1458	1542	634	-634
$\frac{1}{x}M_{ij}^{o}$ (2)	-18000	-18000	3375	3375	7000
	5850	11700	2925	1462	
			-1776	-3551	-8286
$\frac{1}{x}M_{ij}^{c}$ (2)	710	1421	355	177	
			-27	-55	-124
	11	22	5	2	
				0	-2
$\frac{1}{x}M_{ij}$ (2)	-11429	-4857	4857	1410	-1410

7.- Superposición:
$$\begin{cases} M_{AB} = 1297 - 11429x; & M_{BA} = 1458 - 4857x \\ M_{BC} = 1542 + 4857x; & M_{CB} = 634 + 1410x \\ M'_{CD} = -634 - 1410x; & M'_{ED} = 7200 \end{cases}$$

8.- Cálculo de la x: Ecuación de Equilibrio:



$\sum Tv = 0;$

$$(M_{AB} + M_{BA} + 4000)\alpha + (M_{BC} + M_{CB})(\frac{-3}{4}\alpha) + (M'_{CD})(-\alpha) - 2400(2\alpha) - 1600(\frac{3}{2}\alpha) = 0$$

$$(6755 - 16286x) + (2176 + 6267x)(\frac{-3}{4}) + (634 + 1410x) - 4800 - 2400 = 0$$

$$6755 - 16286x - 1632 - 4700,25x - 6566 + 1410x = 0$$

$$-19576,25x - 1443 = 0$$

$$x = -7,371x10^{-2}$$

9.- Momentos Definitivos: $\begin{cases} M_{AB} = 2139; & M_{BA} = 1816 \\ M_{BC} = 1184; & M_{CB} = 530 \\ M'_{CD} = -530; & M'_{ED} = 7200 \end{cases}$

10.- Comprobación: $\begin{cases} M^{c}_{AB} = 1004; & M^{c}_{BA} = 681 \\ M^{c}_{BC} = 454; & M^{c}_{CB} = 1400 \\ M'^{c}_{CD} = 3520; & M'^{c}_{ED} = 0 \end{cases}$

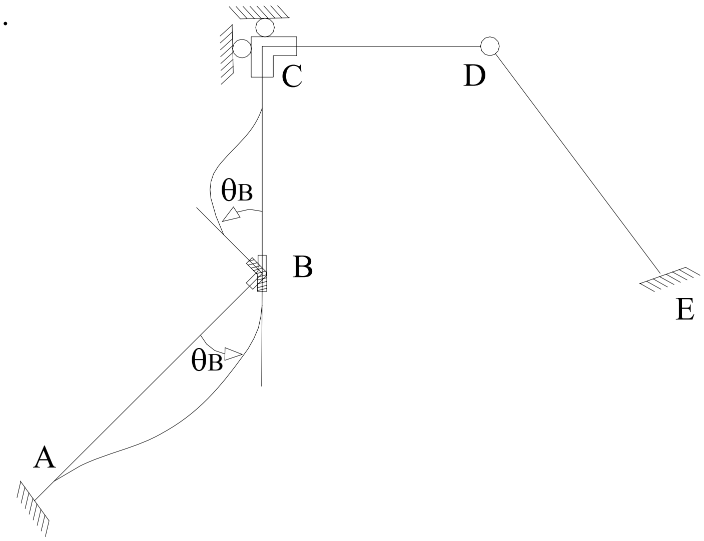
$$\psi_{AB} = \frac{-73,71}{Ekr} \quad ; \quad \psi_{CD} = \frac{73,71}{Ekr} \quad ; \quad \psi_{CB} = \frac{55,28}{Ekr}$$

$$\theta_{CD} = \frac{3520}{7EKr} + \frac{73,71}{EKr} = \frac{576,71}{EKr} \quad ; \quad \theta_{CB} = \frac{1400 - \frac{1}{2}(454)}{3E(\frac{3}{4}Kr)} + \frac{55,28}{EKr} = \frac{576,61}{EKr}$$

Parte Complementaria del Método de las Rotaciones.

Problema Complementario # 1.

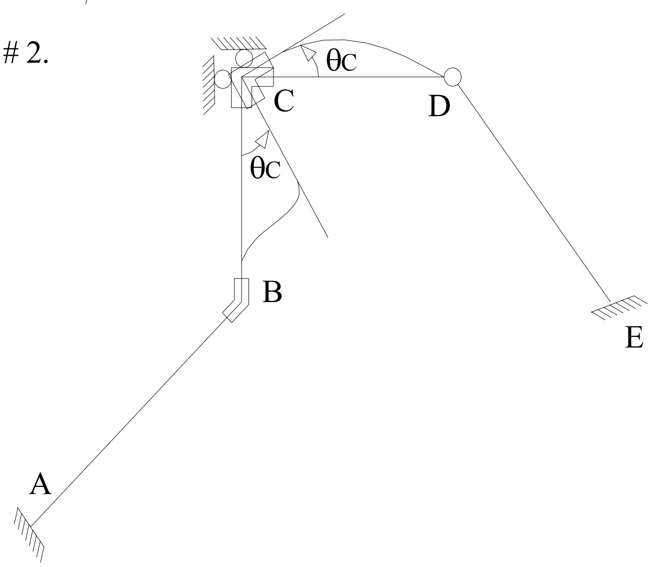
$(\exists \theta_B) \qquad \psi_{ij} = 0$



$$\begin{cases} M'_{CD} = M'_{ED} = 0 \\ M_{BC} = 3EKr\theta_B = 3x \\ M_{CB} = 1{,}5EKr\theta_B = 1{,}5x \\ M_{BA} = 12EKr\theta_B = 12x \\ M_{AB} = 6EKr\theta_B = 6x \end{cases}$$

Problema Complementario # 2.

$(\exists \theta_C) \qquad (\psi_{ij} = 0)$



$$\begin{cases} M_{AB} = M_{BA} = 0 \\ M_{CB} = 3EKr\theta_C = 3y \\ M_{BC} = 1{,}5EKr\theta_C = 1{,}5y \\ M'_{CD} = 7EKr\theta_C = 7y \\ M_{ED} = 0 \end{cases}$$

Problema Complementario # 3 (Etapa de Desplazabilidad).

$(\exists \quad \psi_{ij})$

Se utilizan los valores obtenidos para el cálculo de los ψ_{ij} en el problema # 2 del Método de Cross.

$$\left\{\psi_{AB} = \alpha \right. ; \qquad \left\{\psi_{CD} = -\alpha \right. ; \qquad \left\{\psi_{BC} = \frac{-3}{4}\alpha \right. ;$$

$$\begin{cases} M'_{ED} = 0; \qquad M_{CD} = 7EKr\alpha = 7z \\ M_{CB} = M_{BC} = 3,375EKr\alpha = 3,375z \\ M_{AB} = M_{BA} = -18EKr\alpha = -18z \end{cases}$$

Superposición:
$$\begin{cases} M_{AB} = 6x - 18z + 1135 \\ M_{BA} = 12x - 18z + 1135 \\ M_{BC} = 3x + 1,5y + 3,375z + 730 \\ M_{CB} = 1,5x + 3y + 3,375z - 870 \\ M'_{CD} = 7y + 7z - 4050 \\ M_{ED} = 7200 \end{cases}$$

Ecuaciones de Equilibrio.

Para $\theta_B \to \sum M_B = 0$.

$$M_{AB} + M_{BC} - M_B^{Ext} = 0; \qquad M_B^{Ext} = 3000$$
$$15x + 1,5y - 14,625z + 1865 - 3000 = 0$$
$$15x + 1,5y - 14,625z - 1135 = 0$$

Para $\theta_C \to \sum M_C = 0$.

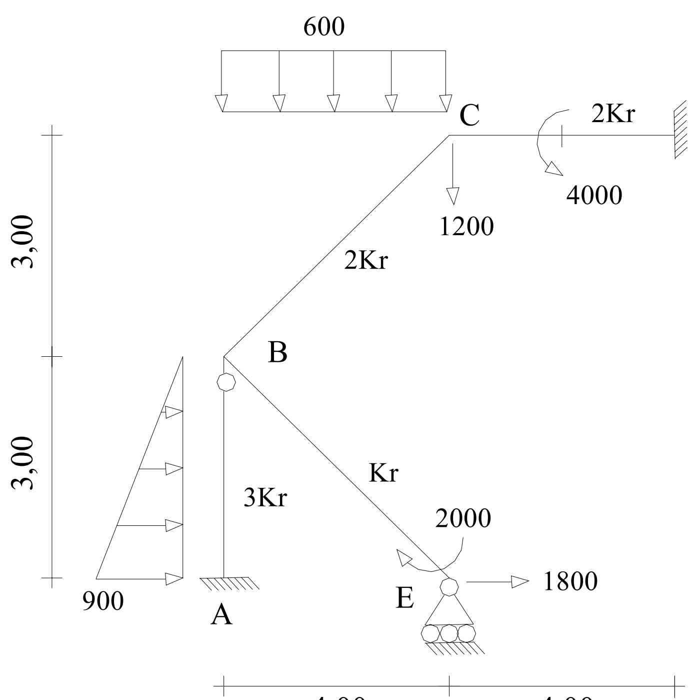
$$M_{CB} + M'_{CD} - M_C^{Ext} = 0; \qquad M_C^{Ext} = 0$$
$$1,5x + 10y + 10,375z - 4920 = 0$$

Para $v_C \to$ Se utilizará la ecuación obtenida en el Trabajo Virtual del Método de Cross para el cálculo de la x.

$$(M_{AB} + M_{BA} + 4000) - 3/4(M_{BC} + M_{CB}) - (M'_{CD}) - 4800 - 2400 = 0$$
$$(18x - 36z + 6270) - 3/4(4,5x + 4,5y + 6,75z - 140) - 7y - 7z + 4050 - 7200 = 0$$
$$18x - 36z + 6270 - 3,375x - 3,375y - 5,0625z + 105 - 7y - 7z - 3150 = 0$$
$$14,625x - 10,375y - 48,0625z + 3225 = 0$$

$$\begin{cases} x = -53,950 \\ y = 576,656 \\ z = -73,796 \end{cases}$$

2.-



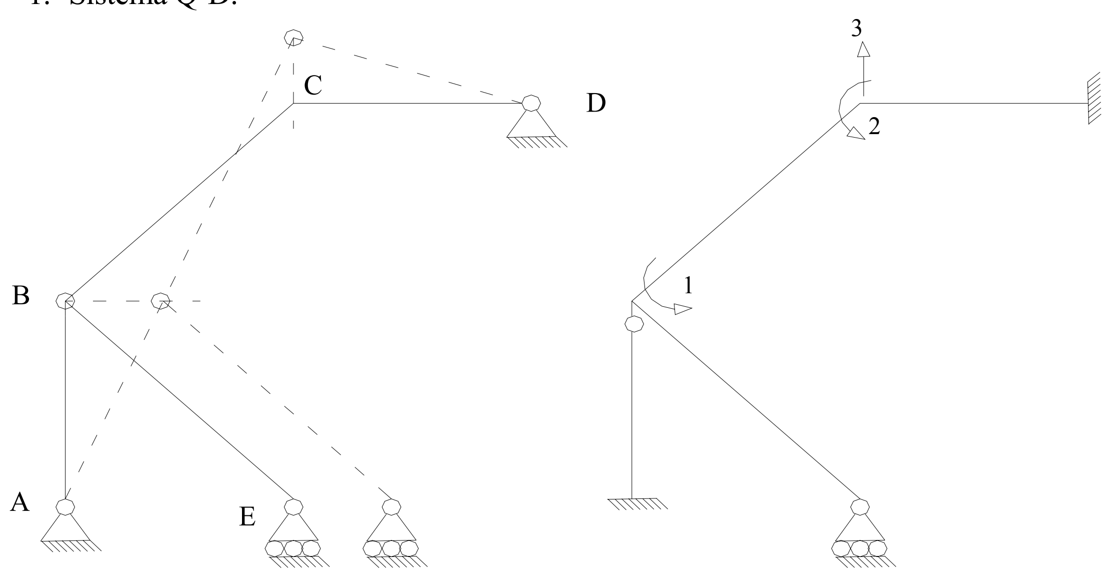
$$\begin{cases} \Delta t_0^{CD} = 10°C \\ \alpha_t = 1x10^{-5}°C^{-1} \\ \mu_D = 2x10^{-2}\,m \\ EKr = 10^5\,Kg.m \end{cases}$$

Determinar:

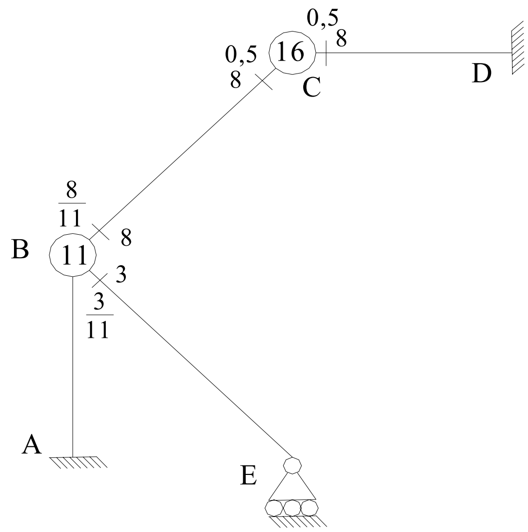
a.- Por el Método de Cross.

b.- Por el Método de las Rotaciones.

1.- Sistema Q-D.



2.- D_{ij} y r_{ij}



3.- Problema #1.

4.- Solución del Problema # 1.

4.1.-Problema Primario: $\theta_B = \theta_C = 0$

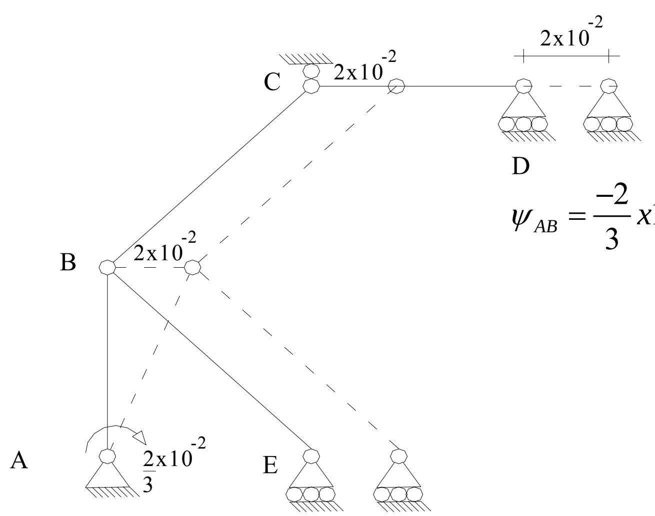
4.1.1.- $M^{E}_{ij} = F$ (cargas, $\Delta(\Delta t)$)

a) F (cargas)

$$\begin{cases} M'^{E}_{AB} = \dfrac{900(3)^2}{20} = 405; & M'^{E}_{BA} = \dfrac{-900(3)^2}{30} = -270 \\ M^{E}_{BC} = \dfrac{600x4^2}{12} = 800; & M^{E}_{CB} = -800 \\ M^{E}_{CD} = \dfrac{4000}{4} = 1000; & M^{E}_{DC} = \dfrac{4000}{4} = 1000 \\ M^{E}_{BE} = M^{E}_{EB} = 0 & \end{cases}$$

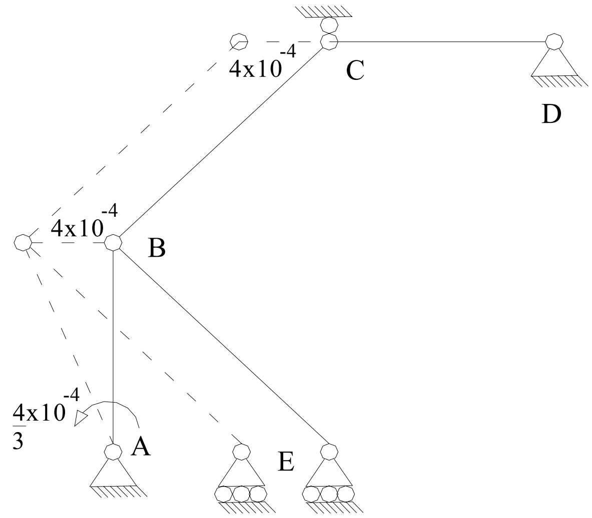
4.1.2.- $\psi^{0}_{ij} =$ F (mov. Apoyo, Δt_o)

F (μ_D)



$$\psi_{AB} = \frac{-2}{3} x10^{-2}$$

F (Δt_o) $\Delta L = 1x10^{-5}x10x4 = 4x10^{-4}$



$\psi_{AB} = \frac{4}{3}x10^{-4}$

Superposición:

$\psi_{AB} = -6,533x10^{-3}$

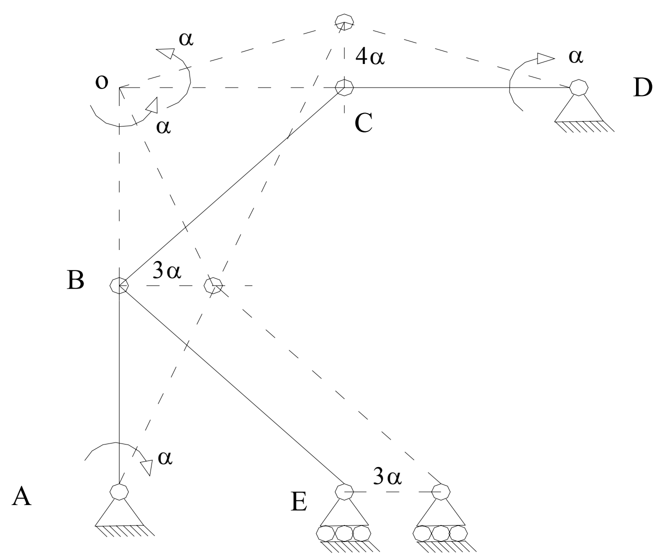
4.1.3.- $M^0_{ij} = ?$

$$\begin{cases} M'^0_{AB} = 405 + 1/2(0+270) + 3E(3Kr)(0+6,533x10^{-3}) = 6419 \\ M'^0_{BE} = 1/2(-2000) = -1000 \\ M^0_{BC} = 800 \quad ; \quad M^0_{CB} = -800 \\ M^0_{CD} = 1000 \quad ; \quad M^0_{DC} = 1000 \end{cases}$$

4.1.4.- $\left\{ M_B^{Fij} = 800 - 1000 = -200; \quad M_C^{Fij} = 1000 - 800 = 200 \right.$

5.- Problema # 2.

5.1.- ψ_{ij} = F (trasl. Apoyo)



$\psi_{AB} = -\alpha$

$\psi_{BC} = \alpha$

$\psi_{DC} = -\alpha$

5.2.- M^o_{ij} = F (ψ_{ij})

$$\begin{cases} M^0_{BC} = M^0_{CB} = -6E(2Kr)(\alpha) = -12EKr\alpha \\ M^0_{CD} = M^0_{DC} = -6E(2Kr)(-\alpha) = 12EKr\alpha \\ M^0_{AB} = 3E(3Kr)(-(-\alpha)) = 9EKr\alpha; \qquad M^0_{BE} = 0 \end{cases}$$

$M^0_{ij}/x = ?$; (EKrα = 1000x)

$$\begin{cases} M^0_{BC}/x = M^0_{CB}/x = -12000; \qquad M^0_{CD}/x = M^0_{DC}/x = 12000 \\ M^0_{AB}/x = 9000; \qquad M^0_{BE}/x = 0 \end{cases}$$

5.3.- $\left\{ M^{Fij}_B / x = -12000; \qquad M^{Fij}_C / x = 12000 - 12000 = 0 \right.$

6.- Solución de las Etapas Complementarias.

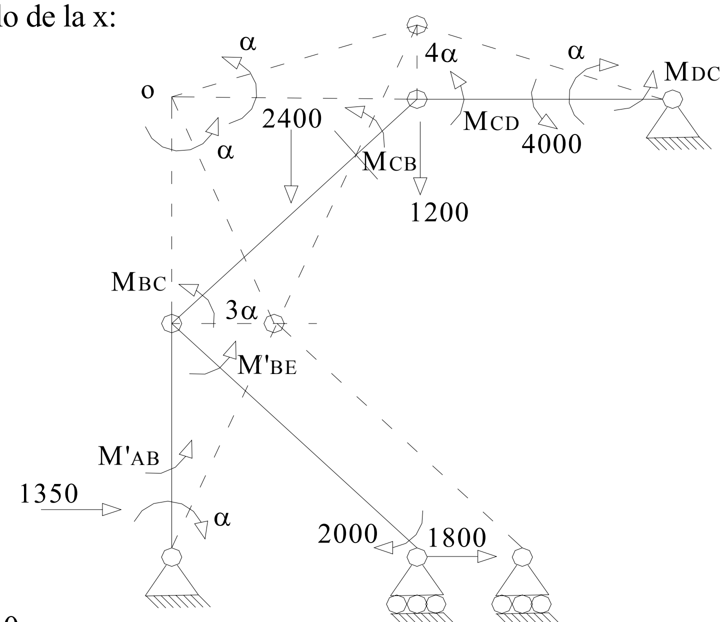
Junta	B		C		D
Miembro	BE	BC	CB	CD	DC
D_{ij}	3/11	8/11	0,5	0,5	0
M^0_{ij} (1)	-1000	800	-800	1000	1000
		-50	-100	-100	-50
	68	182	91		
M^C_{ij} (1)		-22	-45	-45	-23
	6	16	8		
		-2	-4	-4	-2
	0	1			
M_{ij} (1)	-925	925	-850	850	925
$\frac{1}{x}M^o_{ij}$ (2)	0	-12000	-12000	12000	12000
$\frac{1}{x}M^c_{ij}$ (2)	3272	8727	4363		
		-1091	-2182	-2182	-1091
	298	793	396		
		-99	-198	-198	-99
	27	72	36		
		-9	-18	-18	-9
	2	7	3		

		0	-1	-1	0
$\frac{1}{x}M_{ij}$ (2)	3600	-3600	-9600	9600	10800

7.- Superposición: $M_{ij} = M_{ij\,(1)} + M'_{ij\,(2)}$

$$\begin{cases} M'_{BE} = -925 + 3600x; & M_{BC} = 925 - 3600x \\ M_{CB} = -850 - 9600x; & M_{CD} = 850 + 9600x \\ M_{DC} = 925 + 10800x; & M'_{AB} = 6419 + 9000x \end{cases}$$

8.- Cálculo de la x:



$\sum Tv = 0;$

$$(M_{DC} + M_{CD} + 4000)(-\alpha) - 1200(4\alpha) + (M_{CB} + M_{BC})(\alpha) - 2400(2\alpha) +$$
$$+1800(3\alpha) - M'_{BA}\,\alpha + 1350(\alpha) = 0$$

$$-(925{+}10800x{+}850{+}9600x{+}4000) - 4800 + (-850 - 9600x + 925 - 3600x) -$$
$$-4800 + 5400 - 6419 - 9000x + 1350 = 0$$

$$-5775 - 20400x - 4800 + 75 - 13200x + 600 - 6419 - 9000x + 1350 = 0$$

$$-42600x - 14969 = 0$$

$$x = -0{,}3514$$

9.- Momentos Definitivos:
$$\begin{cases} M'_{BE} = -2190; & M_{BC} = 2190 \\ M_{CB} = 2523; & M_{CD} = -2523 \\ M_{DC} = -2870; & M'_{AB} = 3257 \end{cases}$$

10.- Comprobación:

$$\begin{cases} M_{BE}^{c'} = -2190 + 1000 = -1190 \\ M_{BC}^{c} = 2190 - 800 = 1390 \\ M_{CB}^{c} = 2523 + 800 = 3323 \\ M_{CD}^{c} = -2523 - 1000 = -3523 \\ M_{DC}^{c} = -2870 - 1000 = -3870 \\ M_{AB}^{'c} = 3257 - 6419 = -3162 \end{cases}$$

$$\psi_{AB} = \frac{351}{Ekr} \quad ; \quad \psi_{BC} = \frac{-351}{Ekr} \quad ; \quad \psi_{DC} = \frac{351}{Ekr}$$

$$\theta_{BC} = \frac{1390 - \frac{1}{2}(3323)}{3E(2Kr)} - \frac{351}{EKr} = \frac{-396}{EKr} \quad ; \quad \theta_{BE} = \frac{-1190}{3E(Kr)} = \frac{-396}{EKr}$$

Parte Complementaria para el Método de las Rotaciones.

1.- Problema Complementario # 1.

$(\exists \theta_B) \qquad M_{ij} = F(\theta_B)$

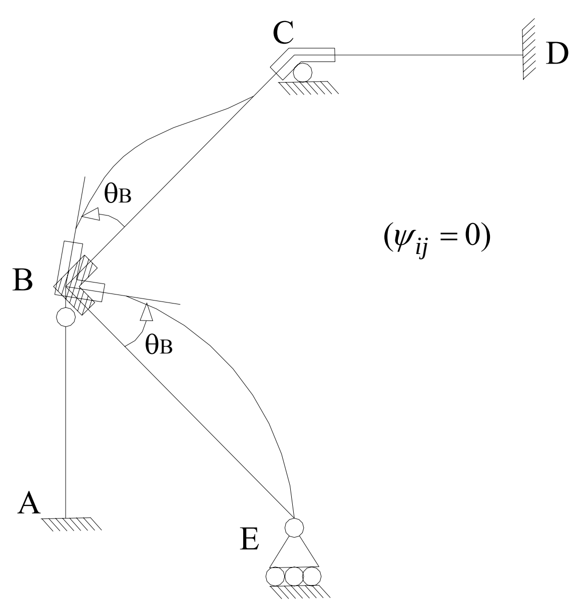
$M_{AB} = 0$

$M_{BC} = 4E(2Kr)\theta_B = 8EKr\theta_B$

$M_{CB} = 2E(2Kr)\theta_B = 4EKr\theta_B$

$M_{CD} = M_{DC} = 0$

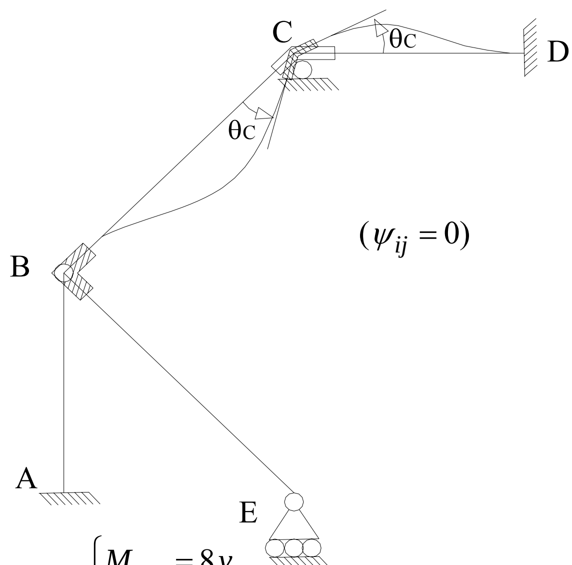
$M_{BE} = 3EKr\theta_B$



Sea $\quad EKr\theta_B = x$

$$\begin{cases} M_{BC} = 8x \\ M_{CB} = 4x \end{cases} \quad ; \quad \begin{cases} M_{BE} = 3x \\ M_{CD} = M_{DC} = M_{AB} = 0 \end{cases}$$

2.- Problema Complementario # 2. $\quad (\exists\,\theta_C)$



$(\psi_{ij} = 0)$

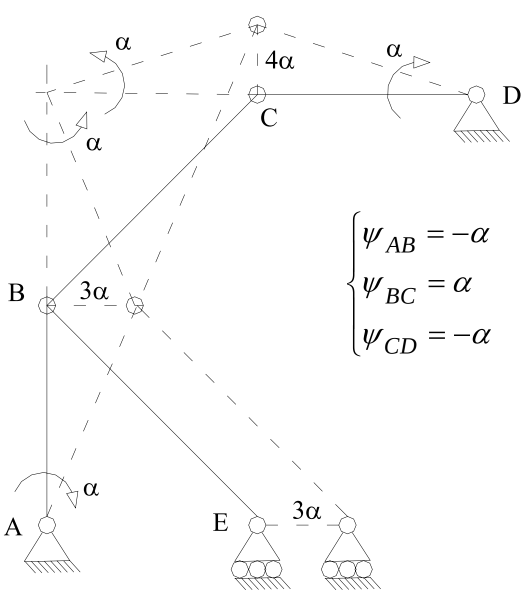
$$\begin{cases} M_{AB} = 0; \qquad M_{BE} = 0 \\ M_{BC} = 2E(2Kr)\theta_C = 4EKr\theta_C \\ M_{CB} = 4E(2Kr)\theta_C = 8EKr\theta_C \\ M_{CD} = 4E(2Kr)\theta_C = 8EKr\theta_C \\ M_{DC} = 4E(Kr)\theta_C = 4EKr\theta_C \end{cases}$$

Sea $\quad EKr\theta_C = y \qquad \begin{cases} M_{AB} = M_{BE} = 0 \\ M_{BC} = 4y \end{cases} \quad ; \quad \begin{cases} M_{CB} = 8y \\ M_{CD} = 8y \\ M_{DC} = 4y \end{cases}$

3.- Problema Complementario # 3 (Etapa de Desplazabilidad).

$(\exists \quad v_C) \qquad (\theta_B = \theta_C = 0)$

Imagen Cinemática:



$$\begin{cases} \psi_{AB} = -\alpha \\ \psi_{BC} = \alpha \\ \psi_{CD} = -\alpha \end{cases}$$

$$M_{ij} = F(\alpha) \qquad \begin{cases} M_{BC} = M_{CB} = -12EKr\alpha; & M_{CD} = M_{DC} = 12EKr\alpha \\ M_{AB} = 9EKr\alpha; & M_{BE} = 0 \end{cases}$$

Sea $\quad EKr\alpha = z$

$$\begin{cases} M_{AB} = 9z; & M_{BC} = M_{CB} = -12z \\ M_{BE} = 0; & M_{CD} = M_{DC} = 12z \end{cases}$$

4.- Superposición:

$$\begin{cases} M_{BC} = 8x + 4y - 12z + 800 \\ M_{CB} = 4x + 8y - 12z - 800 \\ M_{BE} = 3x - 1000 \\ M_{CD} = 8y + 12z + 1000 \\ M_{DC} = 4y + 12z + 1000 \\ M'_{AB} = 9z + 6419 \end{cases}$$

5.- Ecuaciones de Equilibrio.

Para $\theta_B \rightarrow \sum M_B = 0$.

$$M_{BC} + M_{BE} - M_B^{Ext} = 0; \quad M_B^{Ext} = 0$$
$$11x + 4y - 12z - 200 = 0$$

Para $\theta_C \rightarrow \sum M_C = 0$.

$$M_{CB} + M_{CD} - M_C^{Ext} = 0; \quad M_C^{Ext} = 0$$
$$4x + 16y + 200 = 0$$

Para v_B

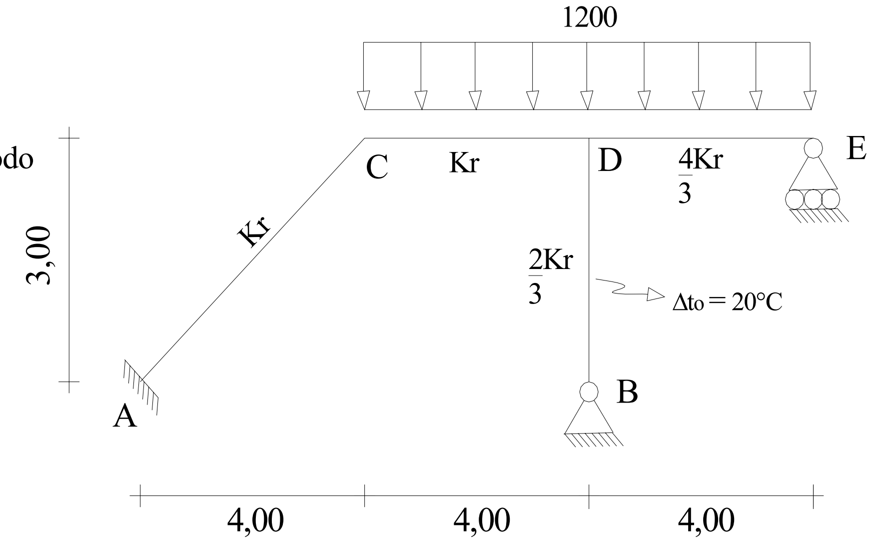
$$-(M_{DC} + M_{CD} + 4000) - 4800 + (M_{CB} + M_{BC}) - 4800 + 5400 - M'_{AB} + 1350 = 0$$
$$-(4y + 12z + 1000 + 8y + 12z + 1000 + 4000) - 9600 + (4x + 8y - 12z - 800 + 8x +$$
$$+ 4y - 12z + 800) + 5400 - 9z - 6419 + 1350 = 0$$
$$-12y - 24z - 6000 - 9600 + 12x + 12y - 24z + 5400 - 9z - 6419 + 1350 = 0$$
$$12x - 57z - 15269 = 0$$

$$\begin{cases} x = -396,662 \\ y = 86,665 \\ z = -351,384 \end{cases}$$

3.- $EKr = 2x10^6 Kg.m$

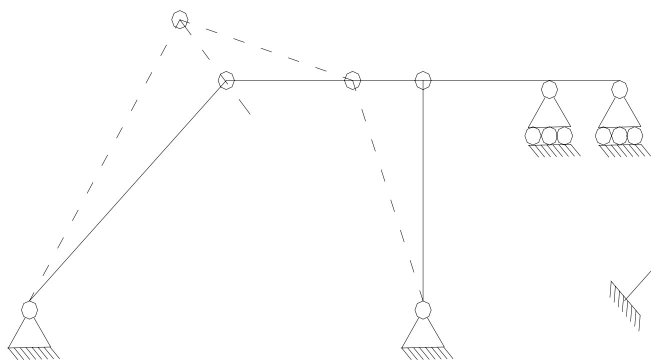
$\alpha t = 1 \times 10^{-5}\ °C^{-1}$

Resolver por el Método de Cross.

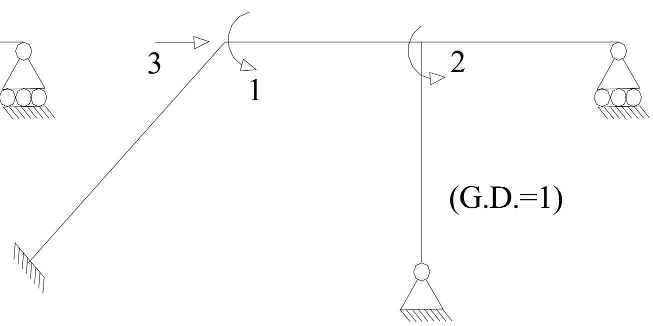


1.- Sistema Q-D.

Imagen Cinemática

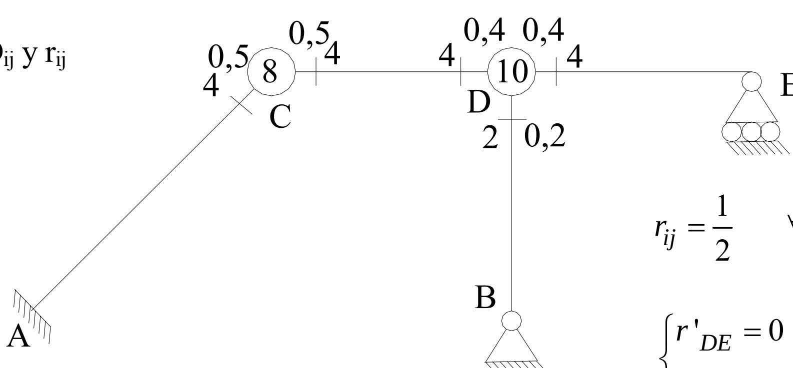


Q-D



2 Sub-problemas

2.- D_{ij} y r_{ij}

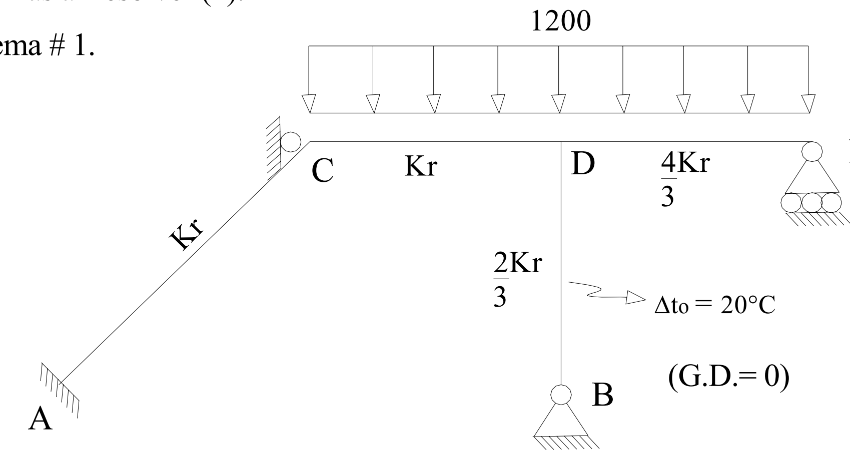


$$r_{ij} = \frac{1}{2} \quad \forall \ ij \neq DE \ y \ DB$$

$$\begin{cases} r'_{DE} = 0 \\ r'_{DB} = 0 \end{cases}$$

3.- Problemas a Resolver (2).

4.- Problema # 1.



4.1.-Problema Primario: $\theta_C = \theta_D = 0$

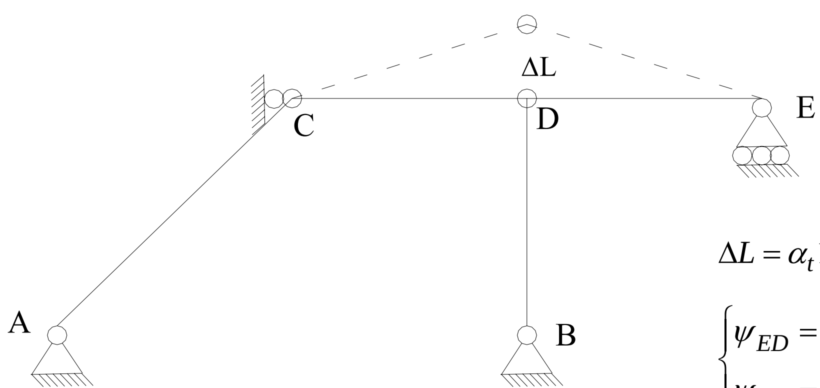
4.1.1.- M^E_{ij} = F (cargas, $\Delta(\Delta t)$) $\Delta t = 0$

M^E_{ij} = F (cargas)

$$\begin{cases} M^E_{AC} = M^E_{CA} = 0; \qquad M^E_{DB} = M^E_{BD} = 0 \\ M^E_{CD} = -M^E_{DC} = \dfrac{1200x4^2}{12} = 1600 \\ M^E_{DE} = -M^E_{ED} = \dfrac{1200x4^2}{12} = 1600 \end{cases}$$

4.1.2.- ψ^0_{ij} = F (mov. Apoyo, Δt_o); (mov. Apoyo = 0)

ψ^0_{ij} = F (Δt_o)



$$\Delta L = \alpha_t 10^{-5} L = 6x10^{-4}$$

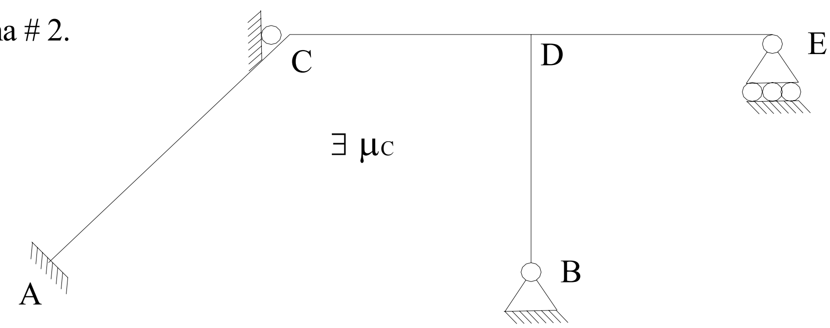
$$\begin{cases} \psi_{ED} = -1,5 \text{ x } 10^{-4} \\ \psi_{CD} = 1,5 \text{ x } 10^{-4} \end{cases}$$

4.1.3.- $M^0_{ij} = ?$ $\qquad \theta_C = \theta_D = 0$

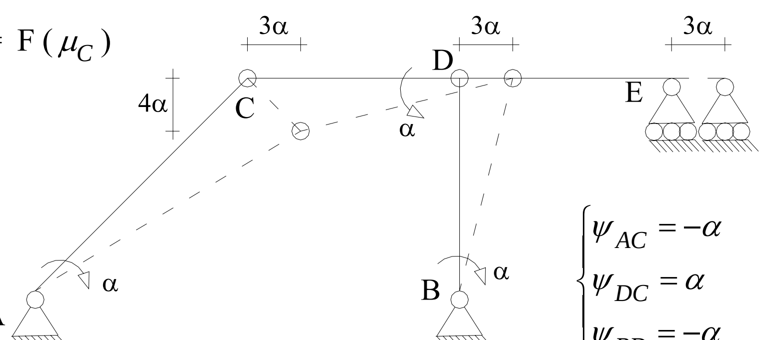
$$\begin{cases} M^0_{AC} = M^0_{CA} = 0; \qquad\qquad M'^0_{DB} = 0 \\ M^0_{CD} = 1600 - 6EKr(1,5x10^{-4}) = 700 \\ M^0_{DC} = -1600 - 6EKr(1,5x10^{-4}) = -2500 \\ M'^0_{DE} = 1600 + \frac{1}{2}(0+1600) + 3E(\frac{4}{3}Kr)(0+1,5x10^{-4}) = 3000 \end{cases}$$

4.1.4.- $M^{Fij}_i = ?$ $\qquad \begin{cases} M^{Fij}_C = 700 \\ M^{Fij}_D = -2500 + 3000 = 500 \end{cases}$

5.- Problema # 2.



5.1.- $\psi^0_{ij} = F(\mu_C)$



$$\begin{cases} \psi_{AC} = -\alpha \\ \psi_{DC} = \alpha \\ \psi_{BD} = -\alpha \end{cases}$$

5.2.- $M^o_{ij} = F(\psi_{ij})$

$$\begin{cases} M_{AC}^{0} = -6EKr(-\alpha) = 6EKr\alpha; & M_{CA}^{0} = -6EKr(-\alpha) = 6EKr\alpha \\ M_{CD}^{0} = -6EKr\alpha; & M_{DC}^{0} = -6EKr\alpha \\ M_{DE}^{0} = 0; & M_{DB}^{0} = 3E(\frac{2}{3}Kr)(-(-\alpha)) = 2EKr\alpha \end{cases}$$

5.2.1.- Cambio de Variable; Sea EKr α = 1000x

$$\begin{cases} M_{AC}^{0}/x = M_{CA}^{0}/x = 6000 \quad ; & M_{DB}^{0}/x = 2000 \\ M_{CD}^{0}/x = M_{DC}^{0}/x = -6000 \quad ; & M_{DE}^{0}/x = 0 \end{cases}$$

5.3.- $\dfrac{M_i^{Fij}}{x} = ?$ $\quad \begin{cases} \dfrac{M_C^{Fij}}{x} = 6000 - 6000 = 0; \end{cases} \quad \dfrac{M_D^{Fij}}{x} = -6000 + 2000 = -4000$

6.- Solución de las Etapas Complementarias:

Junta	A	C			D	
Miembro	AC	CA	CD	DC	DB	DE
D_{ij}	0	0,5	0,5	0,4	0,2	0,4
M_{ij}^{0} (1)	0	0	700	-2500	0	3000
	-175	-350	-350	-175		
			-65	-130	-65	-130
M_{ij}^{C} (1)	16	32	33	16		
			-3	-6	-4	-6
	0	1	2	1		
M_{ij} (1)	-159	-317	317	-2794	-69	2864
$\frac{1}{x}M_{ij}^{o}$ (2)	6000	6000	-6000	-6000	2000	0
$\frac{1}{x}M_{ij}^{c}$ (2)			800	1600	800	1600
	-200	-400	-400	-200		
			40	80	40	80

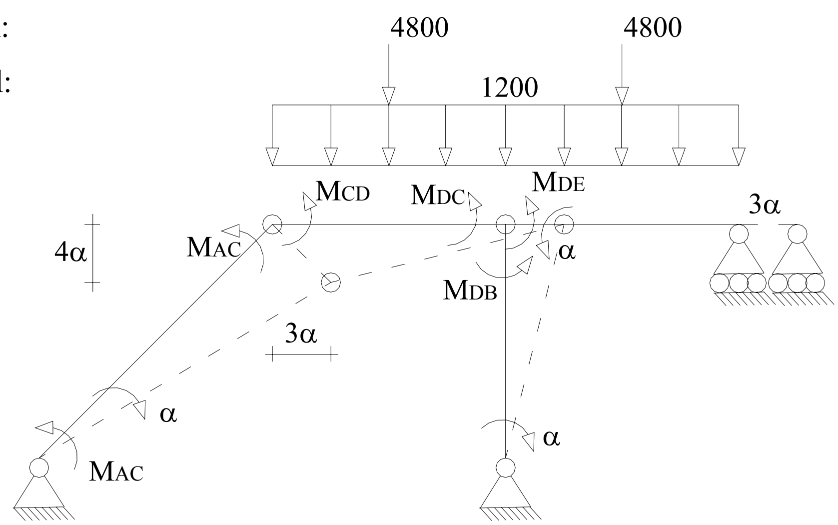
	-10	-20	-20	-10		
			2	4	2	4
	0	-1	-1			
$\frac{1}{x}M_{ij}$ (2)	5790	5579	-5579	-4526	2842	1684

7.- Superposición:

$$\begin{cases} M_{AC} = -159 + 5790x; & M_{CA} = -317 + 5579x \\ M_{CD} = 317 - 5579x; & M_{DC} = -2794 - 4526x \\ M'_{DB} = -69 + 2842x; & M'_{DE} = 2864 + 1684x \end{cases}$$

8.- Cálculo de la x:

Trabajo Virtual:



$$\sum Tv = 0;$$

$$(M_{AC} + M_{CA})(-\alpha) + (M_{CD} + MD_{DC})(\alpha) + (M'_{DB})(-\alpha) + 4800(2\alpha) = 0$$

$$-(-159 + 5790x - 317 + 5579x) + (317 - 5579x - 2794 - 4526x) - (-69 + 2842x) + 9600 = 0$$

$$476 - 11369x - 2477 - 10105x + 69 - 2842x + 9600$$

$$7668 - 24316x = 0$$

$$x = 0,3153$$

9.- Momentos Definitivos: $\begin{cases} M_{AC} = 1667; & M_{CA} = 1442 \\ M_{CD} = -1442; & M_{DC} = -4221 \\ M'_{DB} = 827; & M'_{DE} = 3395 \end{cases}$

10.- Comprobación:

Cálculo de los momentos complementarios. $M_{ij}^{C} = M_{ij} - M_{ij}^{o}$

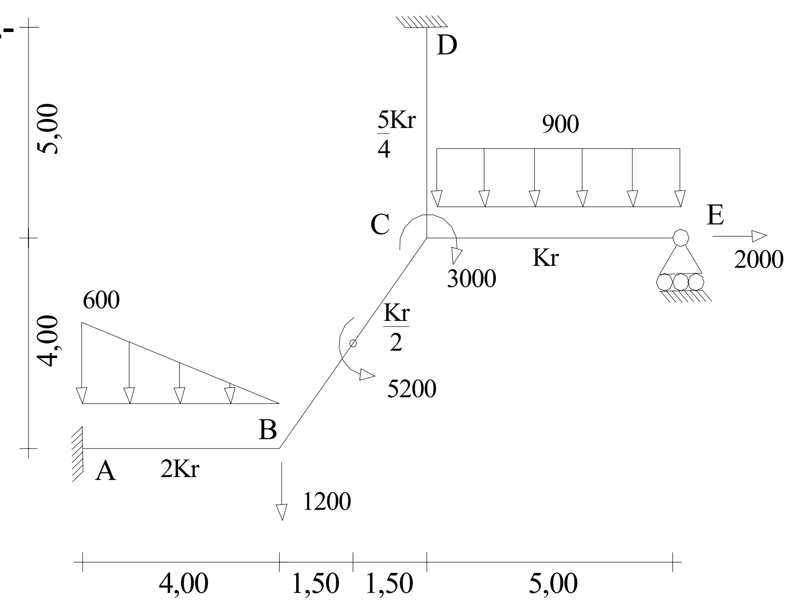
$$\begin{cases} M_{AC}^{c} = 1667 - 0 = 1667; & M_{CA}^{c} = 1442 - 0 = 1442 \\ M_{CD}^{c} = -1442 - 700 = -2142; & M_{DC}^{c} = -4221 + 2500 = -1721 \\ M_{DB}^{'c} = 827 - 0 = 827; & M_{DE}^{'c} = 3395 - 3000 = 395 \end{cases}$$

$$\psi_{ij} = ? \qquad EKr\alpha = 1000x \quad \rightarrow \quad \alpha = \frac{1000x}{EKr} = \frac{315}{EKr}$$

$$\psi_{AC} = \frac{-315}{Ekr} \quad ; \quad \psi_{DC} = \frac{315}{Ekr} \quad ; \quad \psi_{BD} = \frac{-315}{Ekr}$$

$$\theta_{CA} = \frac{1442 - \frac{1}{2}(1667)}{3EKr} - \frac{315}{EKr} = -112,67 \quad ; \quad \theta_{CD} = \frac{-2142 - \frac{1}{2}(-1721)}{3EKr} + \frac{315}{EKr} = -112,167$$

4.-



$\theta_D = 3 \times 10^{-3}$ rad

$\nu_D = -2 \times 10^{-2}$m

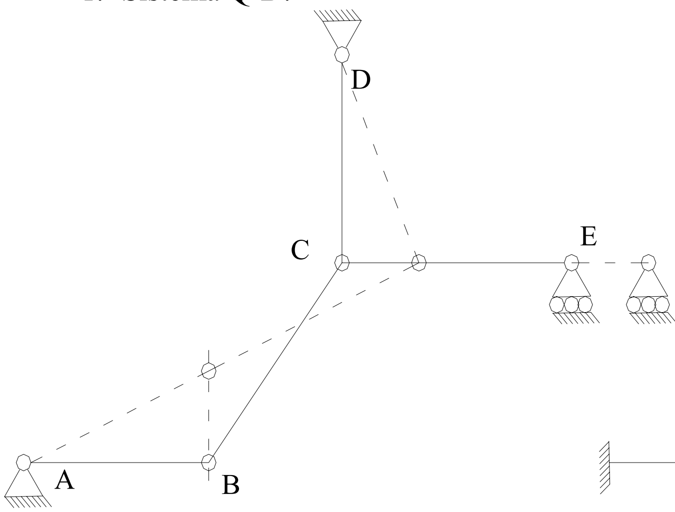
EKr = 10^5Kg.m

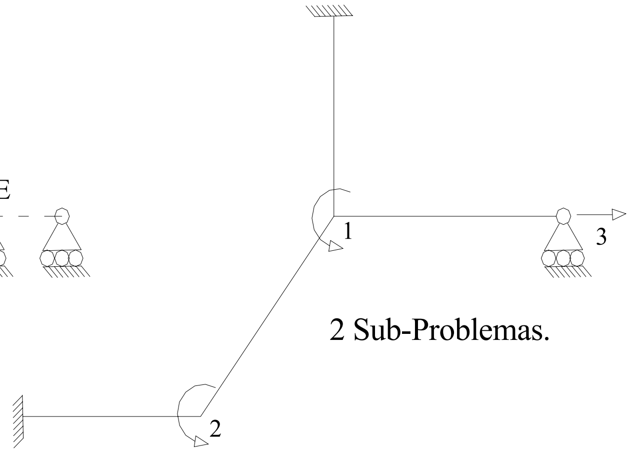
Resolver:

a.- Método de Cross.

b.- Método de las Rotaciones.

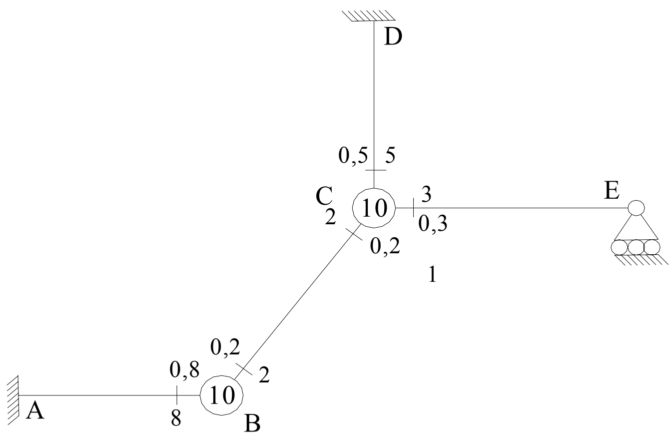
1.- Sistema Q-D.





2 Sub-Problemas.

2.- D_{ij} y r_{ij}



3.- Problema # 1.

4.- Solución:

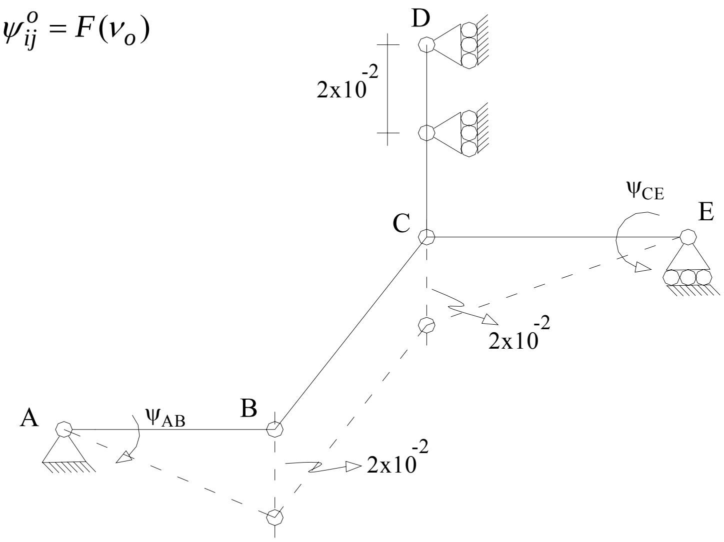
4.1.-Problema Primario: $\theta_B = \theta_C = 0$

4.1.1.- $M^E_{ij} = F$ (cargas)

$$\begin{cases} M^E_{AB} = \dfrac{-600x4^2}{20} = -480; & M^E_{BA} = \dfrac{600x4^2}{30} = 320 \\ M^E_{BC} = \dfrac{5200}{4} = 1300; & M^E_{CB} = 1300 \\ M^E_{CD} = 0; & M^E_{DC} = 0 \\ M^E_{CE} = \dfrac{900x5^2}{12} = 1875; & M^E_{EC} = -1875 \end{cases}$$

4.1.2.- ψ^0_{ij} = F (mov. Apoyo, Δt_o) $\Delta t_o = 0$

$\psi_{ij}^{o} = F(\nu_o)$



$$\begin{cases} \psi_{CE} = \frac{2}{5} x10^{-2} \\ \psi_{AB} = -5 \text{ x } 10^{-3} \end{cases}$$

4.1.3.- M^0_{ij} = ?

$$\begin{cases} M^0_{BA} = 320 - 6E(2Kr)(-5x10^{-3}) = 6320 \\ M^0_{AB} = -480 + 6000 = 5520 \\ M^0_{BC} = 1300; \qquad\qquad M^0_{CB} = 1300 \\ M^0_{CD} = 2E(\frac{5}{4}Kr)(3x10^{-3}) = 750 \\ M^0_{DC} = 4E(\frac{5}{4}Kr)(3x10^{-3}) = 1500 \\ M'^0_{CE} = 1875 + \frac{1}{2}(0 + 1875) + 3EKr[-(\frac{2}{5}x10^{-2})] = 1613 \end{cases}$$

4.1.4.- $M_i^{Fij} = ?$

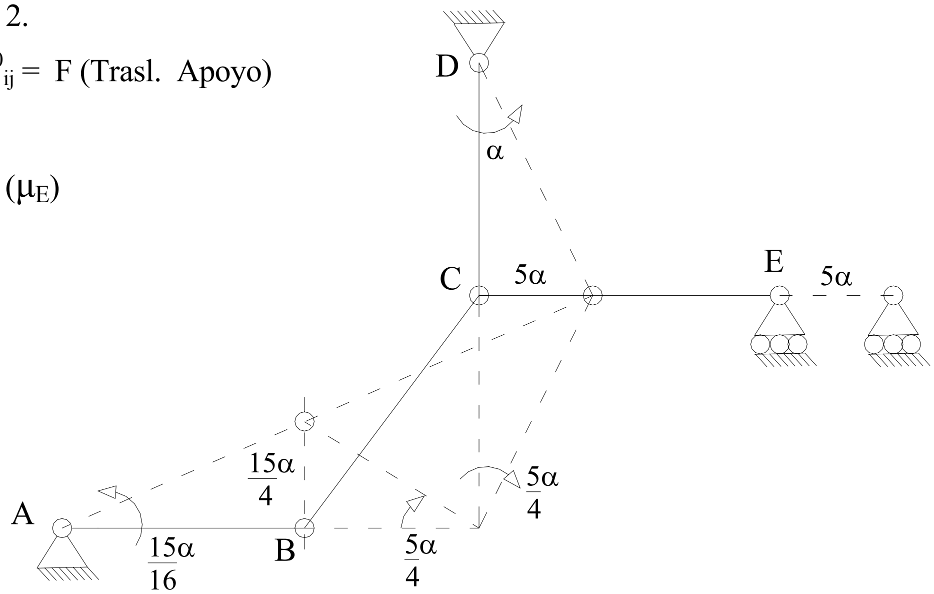
$$\begin{cases} M_B^{Fij} = 6320 + 1300 = 7620 \\ M_C^{Fij} = 1300 + 750 + 1613 + 3000 = 6663 \end{cases}$$

5.- Problema # 2.

5.1.- ψ^0_{ij} = F (Trasl. Apoyo)

ψ_{ij} = F (μ_E)

$$\begin{cases} \psi_{AB} = \dfrac{15}{16}\alpha \\ \psi_{DC} = \alpha \\ \psi_{BC} = \dfrac{-5}{4}\alpha \end{cases}$$



5.2.- M^o_{ij} = F (ψ_{ij})

$$\begin{cases} M^0_{BA} = M^0_{AB} = -6E(2Kr)(\dfrac{15}{16}\alpha) = -11,25EKr\alpha \\ M^0_{BC} = M^0_{CB} = -6E(\dfrac{Kr}{2})(\dfrac{-5}{4}\alpha) = \dfrac{15}{4}EKr\alpha \\ M^0_{CD} = M^0_{DC} = -6E(\dfrac{5}{4}Kr)\alpha = \dfrac{-15}{2}EKr\alpha \\ M^0_{CE} = M^0_{EC} = 0 \end{cases}$$

Sea EKr α = 1000x

$$\frac{M^0_{BA}}{x} = \frac{M^0_{AB}}{x} = -11250 \quad ; \quad \frac{M^0_{BC}}{x} = \frac{M^0_{CB}}{x} = 3750$$

$$\frac{M^0_{CD}}{x} = \frac{M^0_{DC}}{x} = -7500 \quad ; \quad \frac{M^0_{CE}}{x} = \frac{M^0_{EC}}{x} = 0$$

$$\begin{cases} \dfrac{M^{Fij}_B}{x} = -7500 \\ \dfrac{M^{Fij}_C}{x} = -3750 \end{cases}$$

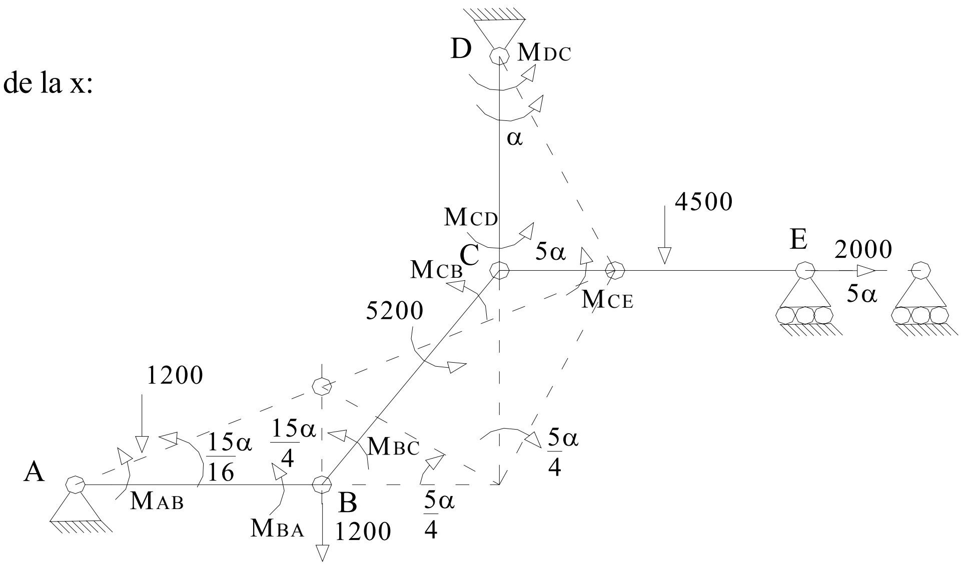
6.- Solución de las Etapas Complementarias.

Junta	A	B		C			D
Miembro	AB	BA	BC	CB	CE	CD	DC
D_{ij}	0	0,8	0,2	0,2	0,3	0,5	0
M_{ij}^{0} (1)	5520	6320	1300	1300	1613	750	1500
	-3048	-6096	-1524	-762			
			-590	-1180	-1770	-2950	-1475
M_{ij}^{C} (1)	236	472	118	59			
			-6	-13	-18	-30	-15
	2	5	1	1			
M_{ij} (1)	2710	701	-701	-595	-175	-2230	10
$\frac{1}{x}M_{ij}^{o}$ (2)	-11250	-11250	3750	3750	0	-7500	-7500
	3000	6000	1500	750			
			300	600	900	1500	750
$\frac{1}{x}M_{ij}^{c}$ (2)	-120	-240	-60	-30			
			3	6	9	15	8
	-1	-2	-1	0			
$\frac{1}{x}M_{ij}$ (2)	-8371	-5492	5492	5076	909	-5985	-6742

7.- Superposición:

$$\begin{cases} M_{AB} = 2710 - 8371x; \\ M_{BC} = -701 + 5492x; \\ M_{CE} = -175 + 909x; \\ M_{DC} = 10 - 6742x \end{cases} \qquad \begin{cases} M_{BA} = 701 - 5492x \\ M_{CB} = -595 + 5076x \\ M_{CD} = -2230 - 5985x \end{cases}$$

8.- Cálculo de la x:



$$\sum Tv = 0;$$

$$(M_{AB} + M_{BA})\frac{15}{16}\alpha - (1200)\frac{15}{12}\alpha + (M_{BC} + M_{CB} + 5200)(\frac{-5}{4}\alpha) + (M_{DC} + M_{CD})(\alpha) +$$

$$+2000(5\alpha) - 1200(\frac{15}{4}\alpha) = 0$$

$$(3411\text{-}13863x)\frac{15}{16} - 1500 + (-701 + 5492x - 595 + 5076x + 5200)(\frac{-5}{4}) + (10 - 6742x -$$

$$-2230 - 5985x) + 10000 - 4500 = 0$$

$$3197{,}8 - 12996{,}56x - 1500 - 4880 - 13210x - 2220 - 12727x + 10000 - 4500 = 0$$

$$97{,}8 - 38933{,}56x = 0$$

$$x = 2{,}511x10^{-3}$$

9.- Momentos Definitivos:

$$\begin{cases} M_{AB} = 2689; \\ M_{BC} = -687; \\ M_{CE} = -173; \\ M_{DC} = -7; \end{cases} \qquad \begin{cases} M_{BA} = 687 \\ M_{CB} = -582 \\ M_{CD} = -2246 \end{cases}$$

10.- Comprobación:
$$\begin{cases} M_{AB}^{c} = -2831; \\ M_{BC}^{c} = -1987; \\ M_{CE}^{c} = -1786; \\ M_{DC}^{c} = -1507; \end{cases} \qquad \begin{cases} M_{BA}^{c} = -5633 \\ M_{CB}^{c} = -1882 \\ M_{CD}^{c} = -2996 \end{cases}$$

$$\psi_{AB} = \frac{2,35}{Ekr} \quad ; \quad \psi_{BC} = \frac{-3,13}{Ekr} \quad ; \quad \psi_{DC} = \frac{2,511}{Ekr}$$

$$\theta_{CD} = \frac{-2996 - \frac{1}{2}(-1507)}{3E(\frac{5}{4}Kr)} + \frac{2,511}{EKr} = \frac{-595}{EKr} \quad ; \quad \theta_{CE} = \frac{-1786}{3EKr} = \frac{-595}{EKr}$$

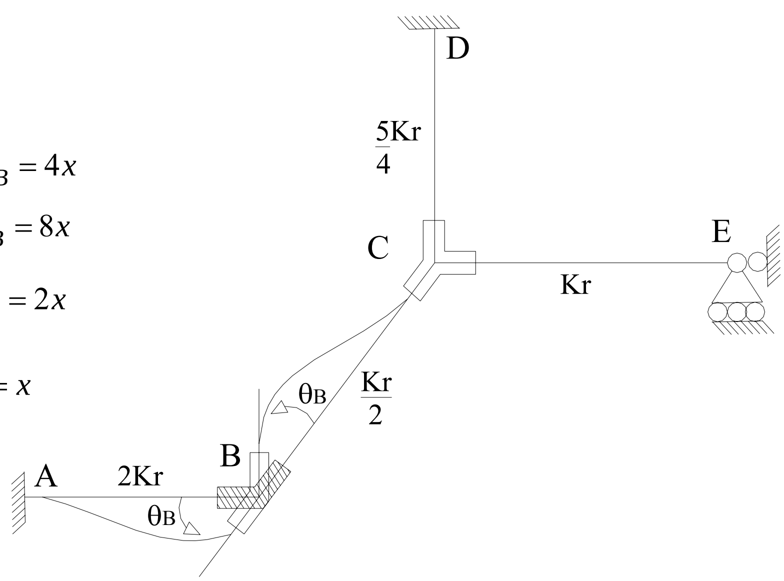
$$\theta_{CB} = \frac{-1882 - \frac{1}{2}(-1987)}{3E(\frac{Kr}{2})} - \frac{3,13}{EKr} = \frac{-595}{EKr}$$

Parte Complementaria del Método de las Rotaciones.

Problema Complementario # 1.

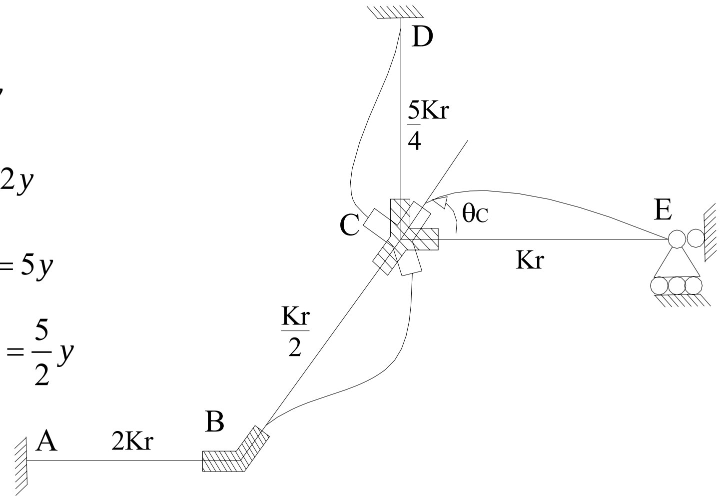
$(\exists \theta_B)$ $\psi_{ij} = 0$

$$\begin{cases} M_{AB} = 2E(2Kr)\theta_B = 4EKr\theta_B = 4x \\ M_{BA} = 4E(2Kr)\theta_B = 8EKr\theta_B = 8x \\ M_{BC} = 4E(\frac{Kr}{2})\theta_B = 2EKr\theta_B = 2x \\ M_{CB} = 2E(\frac{Kr}{2})\theta_B = EKr\theta_B = x \\ M_{CD} = M_{DC} = M'_{CE} = 0 \end{cases}$$



Problema Complementario # 2. $(\exists\theta_C)$

$$\begin{cases} M_{AB}=0; \qquad M_{BA}=0 \\ M_{BC}=2E(\frac{Kr}{2})\theta_C=EKr\theta_C=y \\ M_{CB}=4E(\frac{Kr}{2})\theta_C=2EKr\theta_C=2y \\ M_{CD}=4E(\frac{5}{4}Kr)\theta_C=5EKr\theta_C=5y \\ M_{DE}=2E(\frac{5}{4}Kr)\theta_C=\frac{5}{2}EKr\theta_C=\frac{5}{2}y \\ M'_{CE}=3EKr\theta_C=3y \end{cases}$$



Problema Complementario # 3 (Etapa de Desplazabilidad).

$(\exists\ \mu_E)$ Se utilizan los valores obtenidos para el cálculo de los ψ_{ij} en el problema # 2 del Método de Cross.

$$\psi_{AB}=\frac{15}{16}\alpha; \qquad \psi_{BC}=\frac{-5}{4}\alpha; \qquad \psi_{DC}=\alpha$$

$$\begin{cases} M_{AB}=M_{BA}=-6E(2Kr)\frac{15}{16}\alpha=-11,25EKr\alpha=-11,25z \\ M_{BC}=M_{CB}=-6E(\frac{Kr}{2})(\frac{-5}{4}\alpha)=\frac{15}{4}EKr\alpha=\frac{15}{4}z \\ M_{DC}=M_{CD}=-6E(\frac{5}{4}Kr)\alpha=\frac{-15}{2}EKr\alpha=\frac{-15}{2}z \\ M_{CE}=M_{EC}=0 \end{cases}$$

Superposición:

$$\begin{cases} M_{AB}=4x-11,25z+5520 \\ M_{BA}=8x-11,25z+6320 \\ M_{BC}=2x+y+3,75z+1300 \\ M_{CB}=x+2y+3,75z+1300 \\ M_{CD}=5y-7,5z+750 \\ M_{DC}=2,5y-7,5z+1500 \\ M'_{CE}=3y+1613 \end{cases}$$

Ecuaciones de Equilibrio.

Para $\theta_B \rightarrow \sum M_B = 0$.

$$M_{BA} + M_{BC} - M_B^{Ext} = 0; \qquad M_B^{Ext} = 0$$
$$10x + y - 7{,}5z + 7620 = 0$$

Para $\theta_C \rightarrow \sum M_C = 0$.

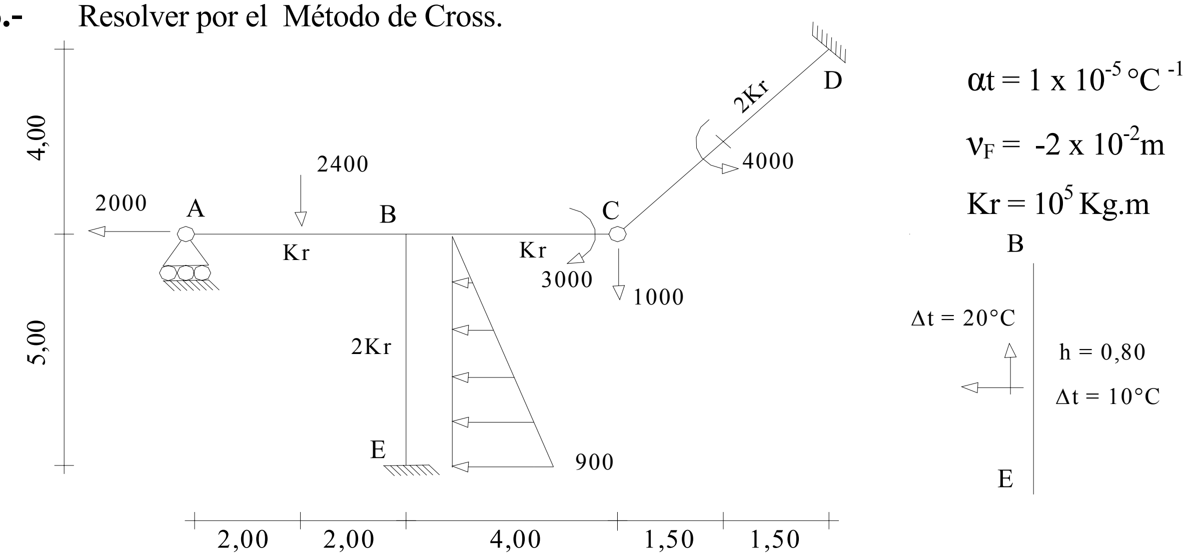
$$M_{CB} + M_{CD} + M'_{CE} - M_C^{Ext} = 0$$
$$x + 10y - 3{,}75z + 6663 = 0$$

Para $\mu_C \rightarrow$ Se utilizará la ecuación planteada en el Tv para el cálculo de la α en el Método de Cross.

$$(M_{AB} + M_{BA})\frac{15}{4} - 1500 + (M_{BC} + M_{CB} + 5200)(\frac{-5}{4}) + (M_{DC} + M_{CD}) + 10000 - 4500 = 0$$
$$(12x - 22{,}50z + 11840)\frac{15}{16} + (3x + 3y + 7{,}5z + 2600 + 5200)(\frac{-5}{4}) + (7{,}5y - 15z + 2250) + 4000 = 0$$
$$11{,}25x - 21{,}09z + 11100 - 3{,}75x - 3{,}75y - 9{,}375z - 9750 + 7{,}5y - 15z + 6250 = 0$$
$$7{,}5x + 3{,}75y - 45{,}475z + 7600 = 0$$

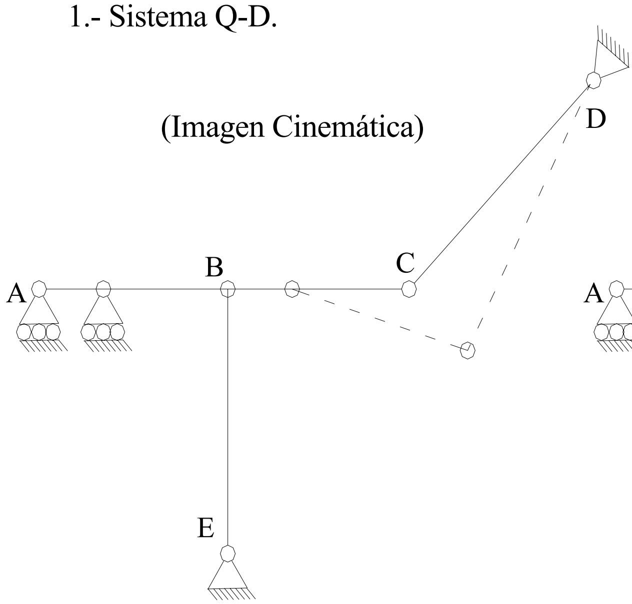
$$\begin{cases} x = -700{,}6045 \\ y = -595{,}3071 \\ z = 2{,}4862 \end{cases}$$

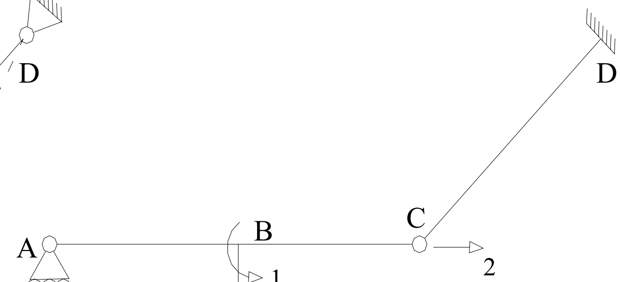
5.- Resolver por el Método de Cross.



1.- Sistema Q-D.

(Imagen Cinemática)

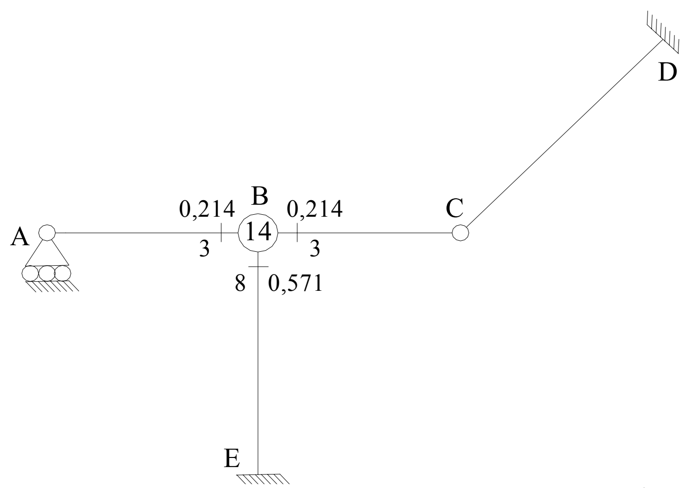




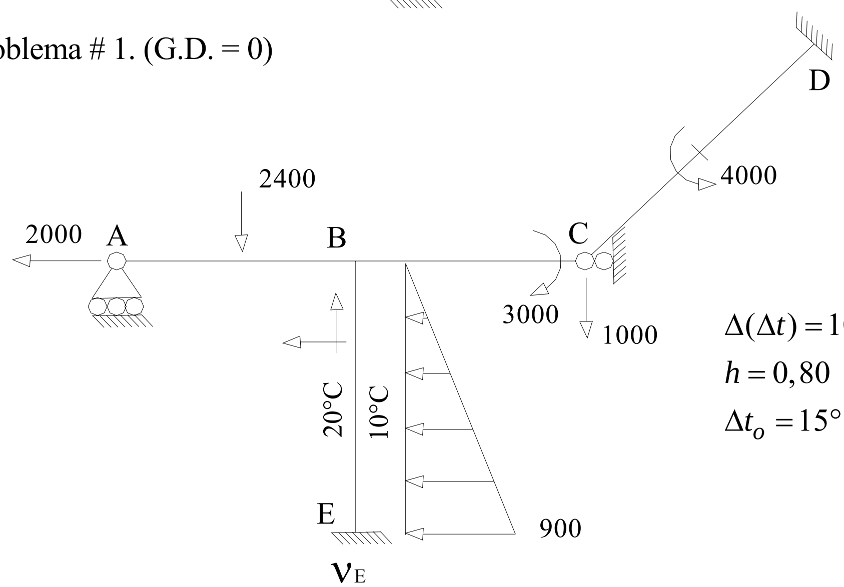
G.D. = 1

2 Sub-Problemas.

2.- D_{ij} y r_{ij}



3.- Problema # 1. (G.D. = 0)



$\Delta(\Delta t) = 10 - 20 = -10$

$h = 0,80$

$\Delta t_o = 15°$

4.- Solución del Problema # 1:

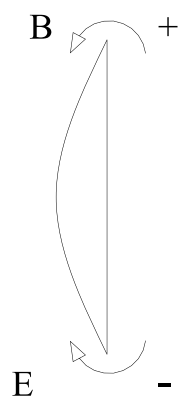
4.1.-Problema Primario: $\theta_B = 0$

4.1.1.- $M^E_{ij} = F$ (cargas, $\Delta(\Delta t)$)

$M^E_{ij} = F$ (cargas)

$$\begin{cases} M^E_{AB} = -M^E_{BA} = \dfrac{2400x4}{8} = 1200; & M^E_{BC} = -M^E_{CB} = 0 \\ M^E_{CD} = M^E_{DC} = \dfrac{4000}{4} = 1000; & M^E_{BE} = \dfrac{900x5^2}{30} = 750 \\ M^E_{EB} = \dfrac{-900x5^2}{20} = -1125 & \end{cases}$$

4.1.2.- $M^E_{ij} = F(\Delta(\Delta t))$ $\qquad K = \dfrac{I}{L} \quad \Rightarrow \quad I = KL$



$$M^E_{BE} = \frac{E[2KrL]\alpha t\Delta(\Delta t)}{h}$$

$$M^E_{BE} = \frac{(10^5)(2)(5)(10^{-5})(+10)}{0,80}$$

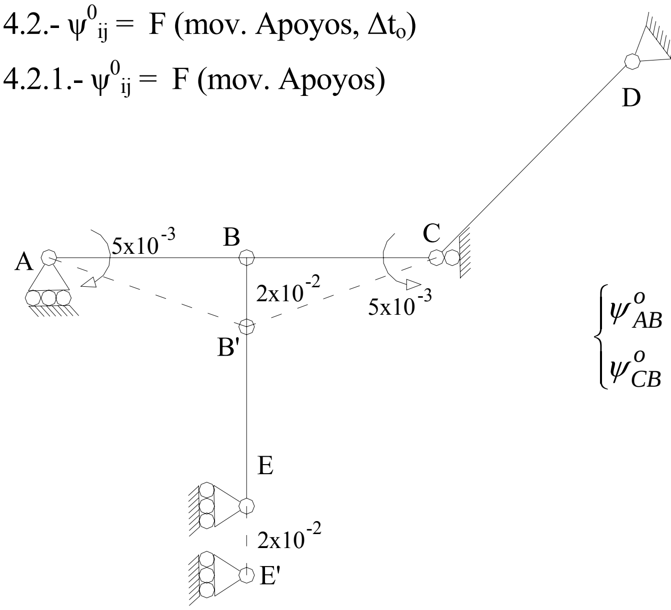
$$M^E_{BE} = 125 = -M^E_{EB}$$

4.1.3.- M^E_{ij} Definitivos.

$$\begin{cases} M^E_{AB} = -M^E_{BA} = 1200 \\ M^E_{CD} = M^E_{DC} = 1000 \end{cases} \qquad \begin{cases} M^E_{BE} = 875 \\ M^E_{EB} = -1250 \end{cases}$$

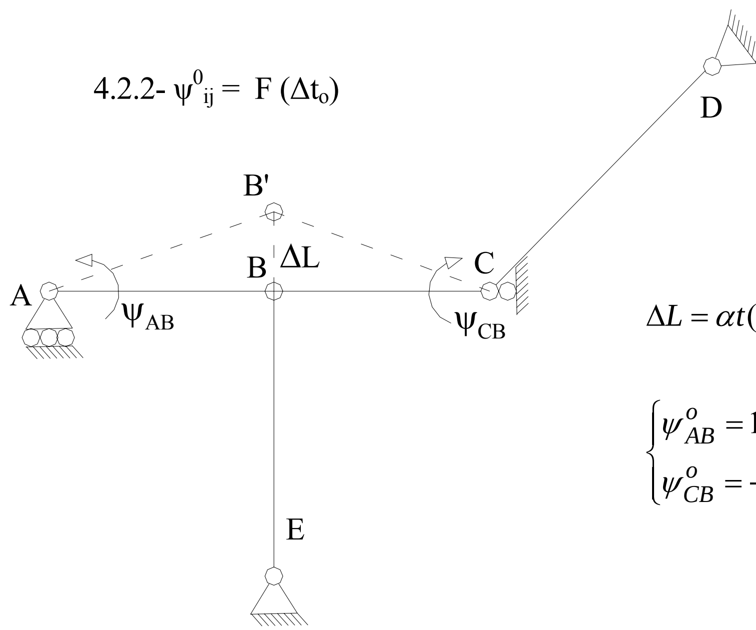
4.2.- $\psi^0_{ij} =$ F (mov. Apoyos, Δt_o)

4.2.1.- $\psi^0_{ij} =$ F (mov. Apoyos)



$$\begin{cases} \psi^o_{AB} = -5x10^{-3} \\ \psi^o_{CB} = 5x10^{-3} \end{cases}$$

4.2.2- $\psi^0_{ij} =$ F (Δt_o)



$$\Delta L = \alpha t(\Delta t)L = 10^{-5}(15)(5) = 7,5x10^{-3}$$

$$\begin{cases} \psi^o_{AB} = 1,875x10^{-4} \\ \psi^o_{CB} = -1875x10^{-4} \end{cases}$$

4.2.3.- Definitivos $\begin{cases} \psi^o_{AB} = -4,81x10^{-3} \\ \psi^o_{CB} = 4,81x10^{-3} \end{cases}$

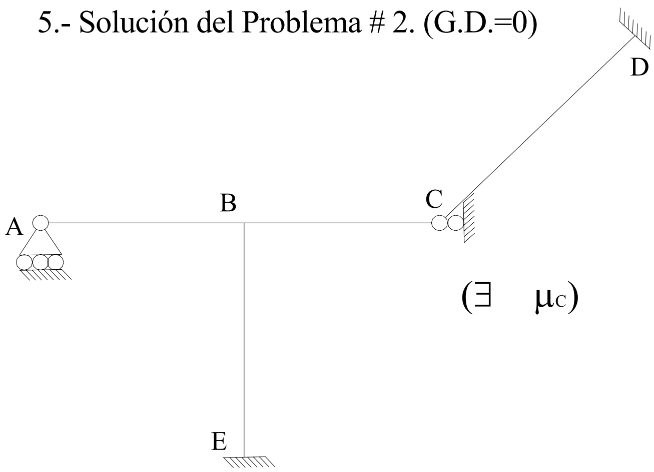
4.3.- $M^0_{ij} = ?$

$$\begin{cases} M'^0_{BA} = -1200 + 1/2(0-1200) + 3EKr(0+4,81x10^{-3}) = -356 \\ M'^0_{BC} = 0 + 1/2(-3000-0) + 3EKr(0-4,81x10^{-3}) = -2944 \\ M'^0_{DC} = 1000 + 1/2(0-1000) = 500 \\ M^0_{BE} = 875; \qquad\qquad M^0_{EB} = -1250 \end{cases}$$

4.4.- $M_i^{Fij} = ?;$ $\qquad M_B^{Ext} = 0$

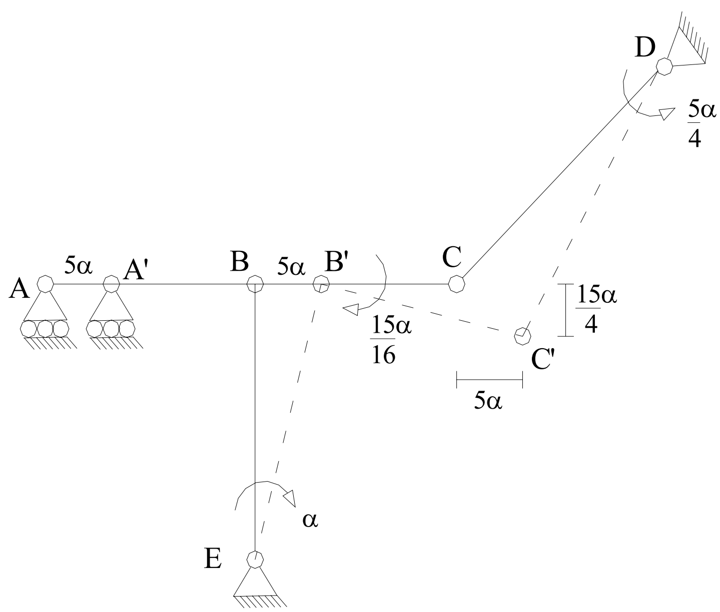
$$M_B^{Fij} = M_{BA}^{o'} + M_{BC}^{o'} + M_{BE}^{o} - M_B^{Ext} = -2425$$

5.- Solución del Problema # 2. (G.D.=0)



5.1.- $\psi_{ij} = F(\mu_C)$

$$\begin{cases} \psi_{BC} = \dfrac{-15}{16}\alpha \\ \psi_{EB} = -\alpha \\ \psi_{DC} = \dfrac{5}{4}\alpha \end{cases}$$



5.2.- $M^{o}_{ij} = F(\psi_{ij})$

$$\begin{cases} M_{BA}^{0'} = 0 \\ M_{BC}^{0'} = 3EKr(\dfrac{15}{16}\alpha) = \dfrac{45}{16}EKr\alpha \\ M_{BE}^{0} = -6E(2Kr)(-\alpha) = 12EKr\alpha \\ M_{EB}^{0} = -6E(2Kr)(-\alpha) = 12EKr\alpha \\ M_{DC}^{0} = 3E(2Kr)(\dfrac{-5}{4}\alpha) = \dfrac{-30EKr\alpha}{4} \end{cases}$$

5.2.1.- Cambio de Variable:

Sea EKr α = 1000x

$\left\{ M_{BC}^{0'} / x = 2812,5 \right.$; $\left\{ M_{BE}^{0} / x = M_{EB}^{0} / x = 12000 \right.$; $\left\{ M_{DC}^{0} / x = -7500 \right.$

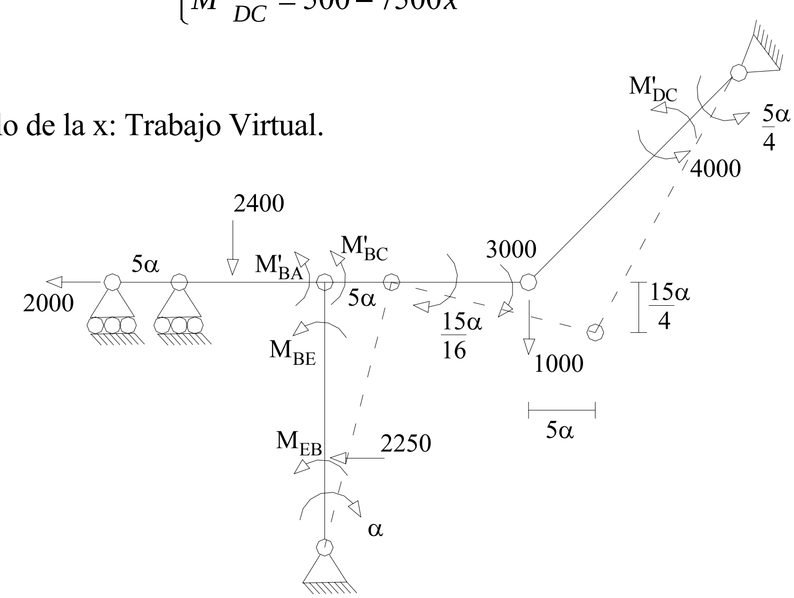
5.3.- $\left\{ M_{B}^{Fij} / x = 2812,5 + 12000 = 14812,5 \right.$

6.-

Junta	B			E
Miembro	BA	BE	BC	EB
D_{ii}	0,214	0,571	0,214	0
M_{ij}^{0} (1)	-356	875	-2944	-1250
	519	1385	519	693
M_{ij} (1)	163	2260	-2425	-557
$\frac{1}{x} M_{ij}^{o}$ (2)	0	12000	2812,5	12000
	-3174	-8464	-3174	-4232
$\frac{1}{x} M_{ij}$ (2)	-3174	3536	-361,5	7768

7.- Superposición: $\begin{cases} M'_{BA} = 163 - 3174x; \\ M'_{BC} = -2425 - 361,5x; \\ M'_{DC} = 500 - 7500x \end{cases}$ $\begin{cases} M_{BE} = 2260 + 3536x \\ M_{EB} = -557 + 7768x \end{cases}$

8.- Cálculo de la x: Trabajo Virtual.



$$\sum Tv = 0;$$

$$(M'_{BC})\frac{-15}{16}\alpha + (M'_{DC})(\frac{5}{4}\alpha) + 4000(\frac{5}{4}\alpha) + 3000(\frac{15}{16}\alpha) + 1000(\frac{15}{4}\alpha) +$$

$$+ (M_{BE} + M_{EB})(-\alpha) - 2000(5\alpha) - 2250(\frac{5}{3}\alpha) = 0$$

$$-(-2425 - 361,5x)\frac{15}{16} + (500 - 7500)(\frac{5}{4}) + 5000 + 2812,5 + 3750 -$$

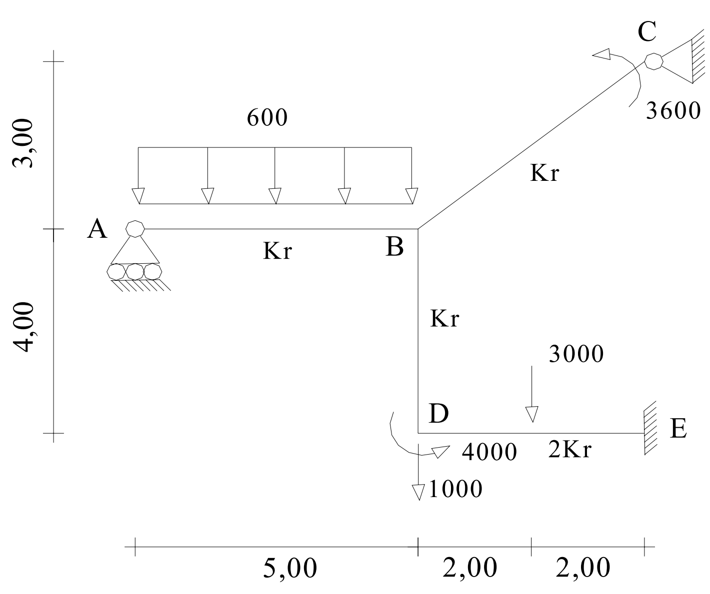
$$-(2260 + 3536x - 557 + 7768x) - 10000 - 3750 = 0$$

$$2273,437 + 338,906x + 625 - 9375x - 2187,5 - 1703 - 11304x = 0$$

$$-992,063 - 20340,094x = 0$$

$$x = -4,8773x10^{-2}$$

9.- Momentos Definitivos: $\begin{cases} M'_{BA} = 317,80 = 318 \\ M_{BE} = 2087,79 = 2088 \\ M'_{BC} = -2407,39 = -2407 \\ M_{EB} = -935,30 = -935 \\ M'_{DC} = 865,25 = 865 \end{cases}$



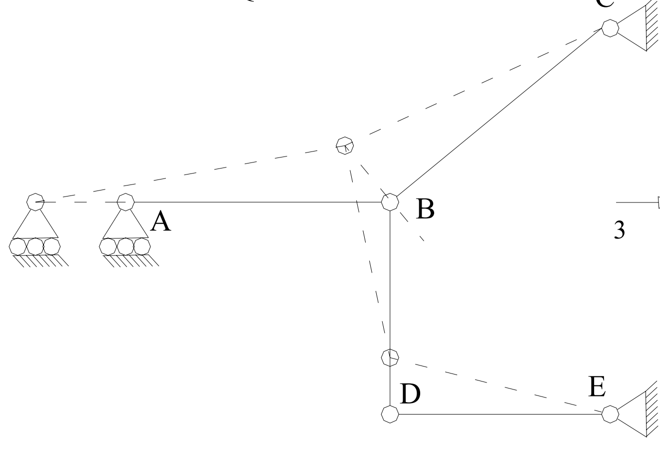
6.- EKr = 10^5 Kg.m

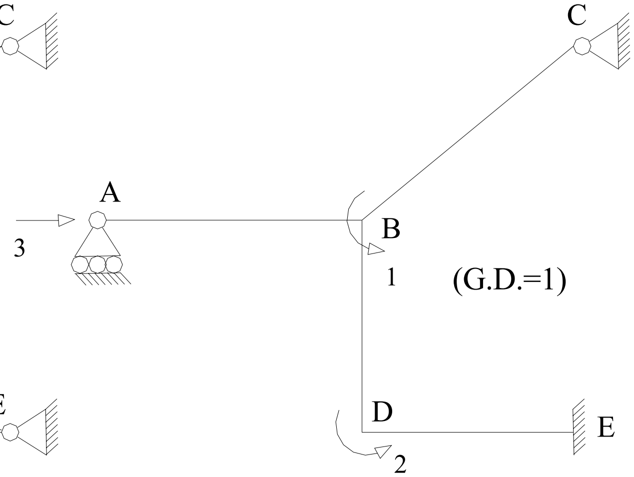
$\Delta t_o^{DE} = 10°C$

$\alpha t = 1 \times 10^{-5}\ °C^{-1}$

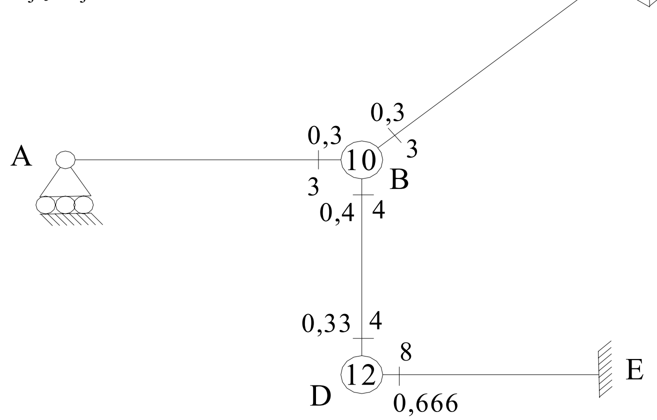
Resolver por el Método de Cross.

1.- Sistema Q-D.





2.- D_{ij} y r_{ij}



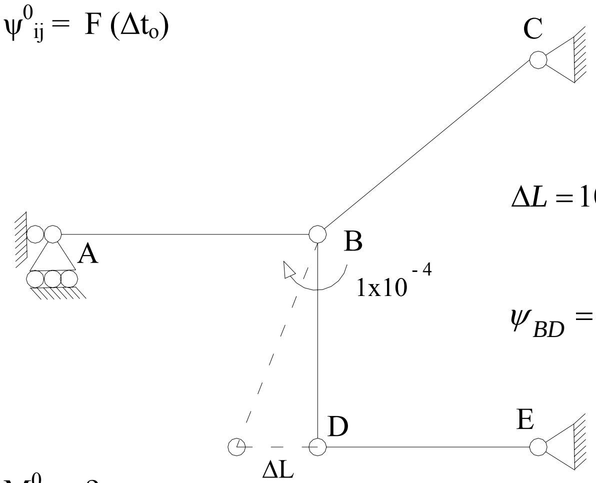
3.- Problema # 1

4.- Problema Primario: $\theta_B = \theta_D = 0$

4.1.- $M^{E}_{ij} = F$ (cargas)

$$\begin{cases} M^{E}_{AB} = -M^{E}_{BA} = \dfrac{600x5^2}{12} = 1250 \\ M^{E}_{BC} = M^{E}_{CB} = 0 \\ M^{E}_{BD} = M^{E}_{DB} = 0 \\ M^{E}_{DE} = -M^{E}_{ED} = \dfrac{3000x4}{8} = 1500 \end{cases}$$

4.2.- $\psi^{0}_{ij} =$ F (Δt_o)



$$\Delta L = 10^{-5} x 10 x 4 = 4 x 10^{-4}$$

$$\psi_{BD} = -1 x 10^{-4}$$

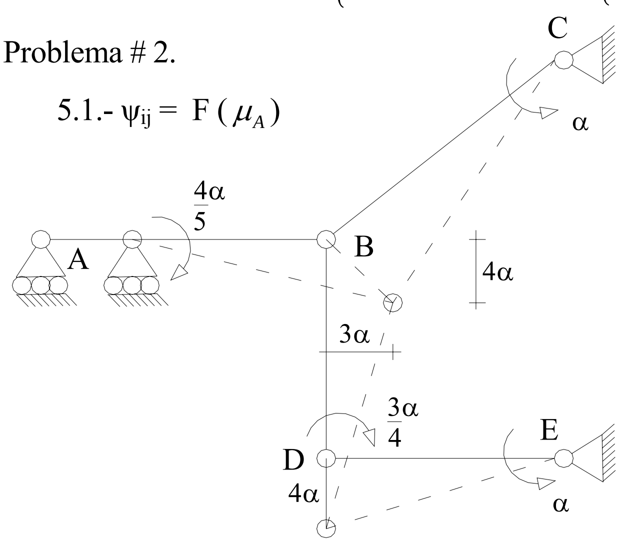
4.3.- $M^{0}_{ij} = ?$

$$\begin{cases} M^{0'}_{BA} = -1250 + 1/2(0 - 1250) = -1875; \\ M^{0}_{BD} = -6EKr(-1x10^{-4}) = 60; \\ M^{0}_{DE} = 1500; \end{cases} \qquad \begin{cases} M^{0'}_{BC} = 1/2(3600) = 1800 \\ M^{0}_{DB} = 60 \\ M^{0}_{ED} = -1500 \end{cases}$$

4.4.- $M^{Fij}_{i} = ?$ $\quad \{ M^{Fij}_{B} = -15; \quad \{ M^{Fij}_{D} = 1650 - 4000 = -2440$

5.- Problema # 2.

5.1.- $\psi_{ij} =$ F (μ_A)



$$\begin{cases} \psi_{AB} = \dfrac{-4}{5}\alpha \\ \psi_{CB} = \alpha \end{cases} \qquad \begin{cases} \psi_{ED} = \alpha \\ \psi_{DB} = \dfrac{-3}{4}\alpha \end{cases}$$

5.2.- $M^{o}_{ij} =$ F (ψ_{ij})

$$\begin{cases} M^{0}_{BA} = 3EKr(-(-4/5\alpha)) = 12/5EKr\alpha \\ M^{0}_{BC} = 3EKr(-\alpha) = -3EKr\alpha \\ M^{0}_{BD} = M^{0}_{DB} = -6EKr(-3/4\alpha) = 4,5EKr\alpha \\ M^{0}_{DE} = M^{0}_{ED} = -6E(2Kr)\alpha = -12EKr\alpha \end{cases}$$

Cambio de Variable: Sea EKr α = 1000x

$$M^{0}_{BA} / x = 2400 \qquad ; \qquad M^{0}_{DE} / x = M^{0}_{ED} / x = -12000$$

$$M^{0}_{BC} / x = -3000 \qquad ; \qquad M^{0}_{BD} / x = M^{0}_{DB} / x = 4500$$

5.3.- $M_i^{Fij} / x = ?$ $\quad \begin{cases} M_B^{Fij} / x = 2400 - 3000 + 4500 = 3900 \\ M_D^{Fij} / x = -12000 + 4500 = -7500 \end{cases}$

6.- Solución de las Etapas Complementarias

Junta		B		D		E
Miembro	BA	BC	BD	DB	DE	ED
D_{ij}	0,3	0,3	0,4	0,333	0,667	0
M^{o}_{ij} (1)	-1875	1800	60	60	1500	-1500
			406	813	1627	813
	-117	-117	-156	-78		
M^{C}_{ij} (1)			13	26	52	26
	-4	-4	-5	-2		
				1	1	
M_{ij} (1)	-1996	1679	318	820	3180	-661
$\frac{1}{x} M^{o}_{ij}$ (2)	2400	-3000	4500	4500	-12000	-12000
			1250	2500	5000	2500
	-1545	-1545	-2060	-1030		
$\frac{1}{x} M^{o}_{ij}$ (2)			172	343	687	343
	-52	-52	-69	-34		
			6	11	22	11
	-1	-1	-2	-1		
$\frac{1}{x} M_{ij}$ (2)	802	-4598	3797	6290	-6290	-9146

Comprobación:

Cálculo de los momentos complementarios: $\quad M^{c}_{ij} = M_{ij} - M^{o}_{ij}$

$$\begin{cases} M_{BA}^c = -1469 + 1875 = 406; \\ M_{BD}^c = 2811 - 60 = 2751; \\ M_{DE}^c = -949 - 1500 = -2449; \end{cases} \qquad \begin{cases} M_{BC}^c = -1340 - 1800 = -3140 \\ M_{DB}^c = 4949 - 60 = 4889 \\ M_{ED}^c = -6665 + 1500 = -5165 \end{cases}$$

$$\psi_{ij} = ?$$

Sea $Ekr\alpha = 1000x \Rightarrow \alpha = \frac{1000x}{EKr} = \frac{656}{EKr}$

$$\psi_{AB} = \frac{-4}{5}\left(\frac{656}{Ekr}\right) = \frac{-525}{Ekr}; \qquad \psi_{CB} = \psi_{ED} = \frac{656}{Ekr}; \qquad \psi_{DB} = \frac{-3}{4}\left(\frac{656}{Ekr}\right) = \frac{-492}{Ekr}$$

$$\theta_{BA} = \frac{406}{3EKr} - \frac{525}{EKr} = \frac{-390}{EKr}; \qquad \theta_{BC} = \frac{-3140}{3EKr} + \frac{656}{EKr} = \frac{-390}{EKr}$$

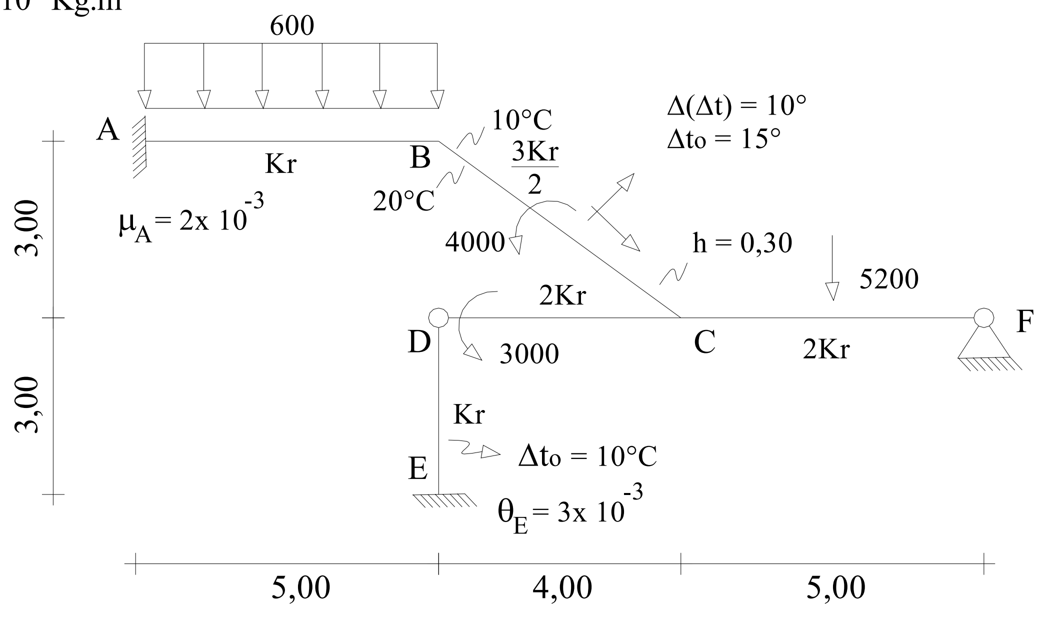
$$\theta_{BD} = \frac{2751 - \frac{1}{2}(4889)}{3EKr} - \frac{492}{EKr} = \frac{-390}{EKr}$$

7.- Resolver:

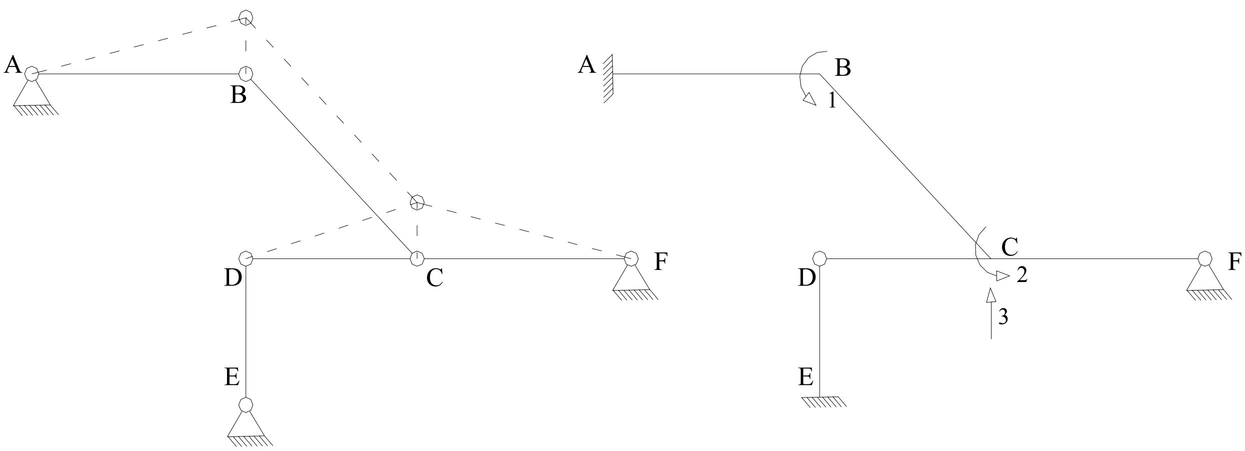
a.- Método de Cross.

b.- Método de las Rotaciones.

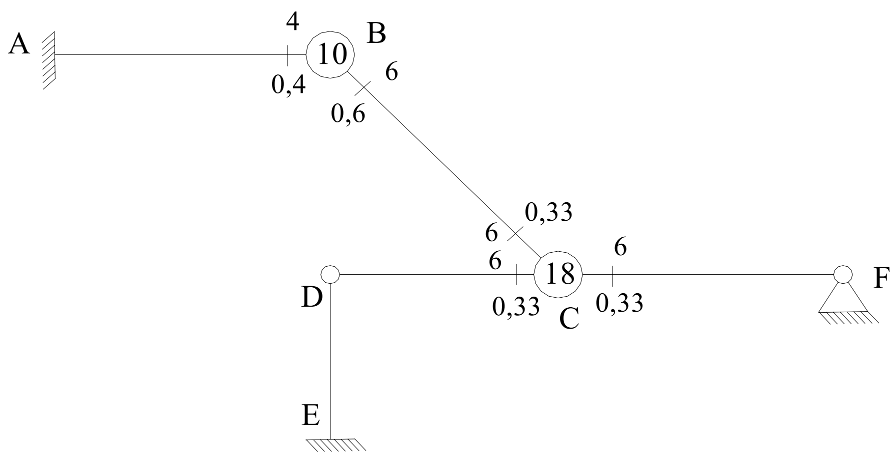
$EKr = 10^5$ Kg.m



1.- Sistema Q –D.



2.- D_{ij} y r_{ij}



3.- Problema # 1.

4.- Solución del Problema # 1.

4.1.- $M_{ij}^{E} = F(\text{cargas}, \Delta(\Delta t))$

a.- F (cargas)

$$\begin{cases} M_{AB}^{E} = 1250; \\ M_{CB}^{E} = 1000; \\ M_{CF}^{E} = 3250; \end{cases} \quad \begin{cases} M_{BA}^{E} = -1250; \\ M_{CD}^{E} = 0; \\ M_{FC}^{E} = -3250; \end{cases} \quad \begin{cases} M_{BC}^{E} = 1000 \\ M_{DC}^{E} = 0 \\ M_{DE}^{E} = M_{ED}^{E} = 0 \end{cases}$$

b.- F (Δ(Δt)).

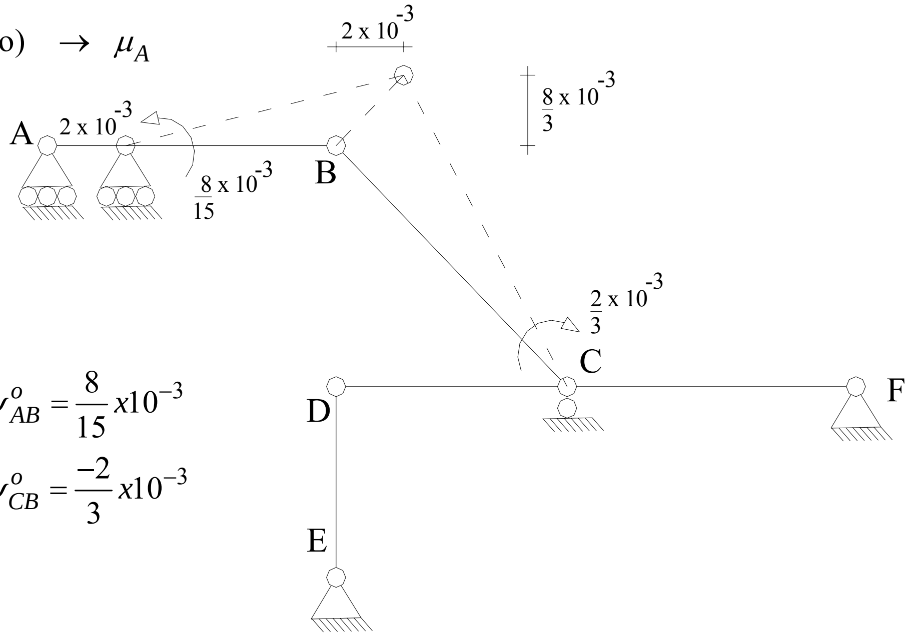
$$M_{BC}^{E}=EKrL\alpha\ t\frac{\Delta(\Delta t)}{h}=\frac{3}{2}x10^{5}x5x10^{-5}x\frac{10}{0,3}=250; \qquad M_{CB}^{E}=-250$$

c.- Superposición

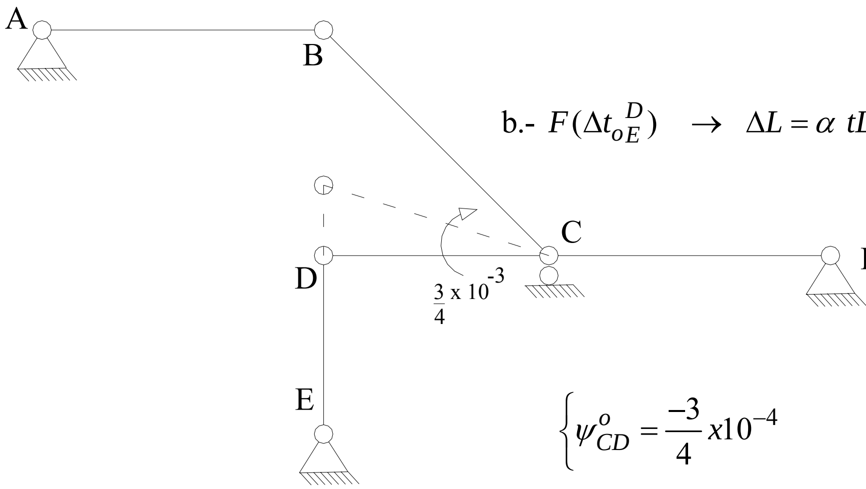
$$\begin{cases} M_{AB}^{E}=-M_{BA}^{E}=1250; & M_{BC}^{E}=1250 \\ M_{CF}^{E}=-M_{FC}^{E}=3250; & M_{CB}^{E}=750 \\ M_{CD}^{E}=M_{DC}^{E}=M_{DE}^{E}=M_{ED}^{E}=0 \end{cases}$$

4.2.- $\psi_{ij}^{o}=F(\text{mov. Apoyo y } \Delta t_{o})$

a.- $F(\text{mov. Apoyo}) \quad \rightarrow \quad \mu_{A}$



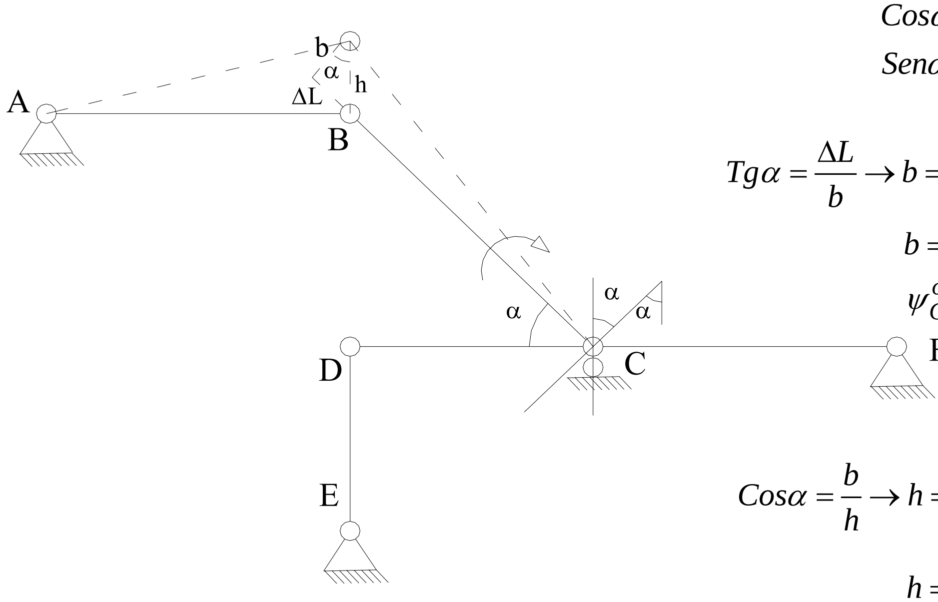
$$\begin{cases} \psi_{AB}^{o}=\dfrac{8}{15}x10^{-3} \\ \psi_{CB}^{o}=\dfrac{-2}{3}x10^{-3} \end{cases}$$



b.- $F(\Delta t_{oE}^{D}) \quad \rightarrow \quad \Delta L=\alpha\ tL\Delta t=10^{-5}x3x10=3x10^{-4}$

$$\left\{\psi_{CD}^{o}=\frac{-3}{4}x10^{-4}\right.$$

c.- $F(\Delta t_{oB}^{C}) \quad \rightarrow \quad \Delta L = \alpha\ tL\Delta t = 10^{-5}x5x15 = 7,5x10^{-4}$



$$Cos\alpha = 4/5$$
$$Sen\alpha = 3/5$$

$$Tg\alpha = \frac{\Delta L}{b} \rightarrow b = \frac{7,5x10^{-4}}{3/4}$$
$$b = 1x10^{-3}$$
$$\psi_{CB}^{o} = -2x10^{-4}$$

$$Cos\alpha = \frac{b}{h} \rightarrow h = \frac{b}{Cos\alpha}$$
$$h = \frac{1x10^{-3}}{4/5}$$
$$h = 1,25x10^{-3}$$
$$\psi_{AB}^{o} = 2,5x10^{-4}$$

c.- Superposición
$$\begin{cases} \psi_{AB}^{o} = 7,83x10^{-4} \\ \psi_{CB}^{o} = -8,67x10^{-4} \\ \psi_{CD}^{o} = -7,5x10^{-5} \end{cases}$$

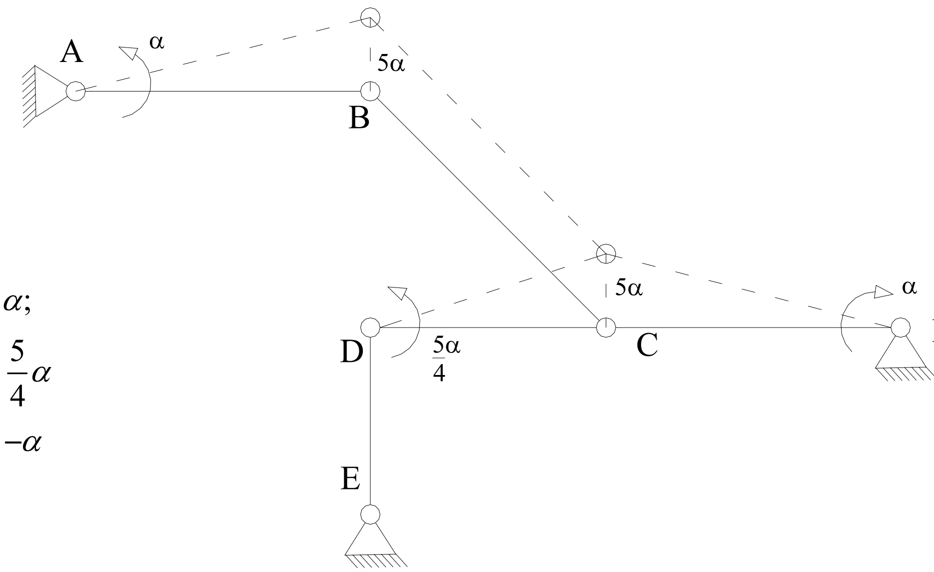
d.- $M_{ij}^{o} = ?$

$$M_{AB}^{o} = 1250 - 6EKr(7,83x10^{-4}) = 780$$
$$M_{BA}^{o} = -1250 - 6EKr(7,83x10^{-4}) = -1720$$
$$M_{BC}^{o} = 1250 - 6E(\frac{3}{2}Kr)(-8,67x10^{-4}) = 2030$$
$$M_{CB}^{o} = 750 - 6E(\frac{3}{2}Kr)(-8,67x10^{-4}) = 1530$$
$$M_{CF}^{o'} = 3250 + \frac{1}{2}(0 + 3250) = 4875$$
$$M_{CD}^{o'} = 0 + \frac{1}{2}(3000 - 0) + 3E(2Kr)(0 + 7,5x10^{-5}) = 1545$$
$$M_{ED}^{o'} = 0 + \frac{1}{2}(0 - 0) + 3E(Kr)(3x10^{-3} - 0) = 900$$

e.- $M^{Fij} = ? \quad \{ M_{B}^{Fij} = 310; \qquad \{ M_{C}^{Fij} = 7950$

5.- Problema # 2.

5.1.- $\psi_{ij} = F(\text{trasl. Apoyo})$ Trabajo Virtual



$$\begin{cases} \psi_{AB} = \alpha; \\ \psi_{DC} = \dfrac{5}{4}\alpha \\ \psi_{CF} = -\alpha \end{cases}$$

5.2.- $M_{ij}^{o} = F(\psi_{ij})$ $\qquad (EKr\alpha = 1000)$

$$\begin{cases} \dfrac{M_{AB}^{o}}{x} = \dfrac{M_{BA}^{o}}{x} = -6E(Kr)\alpha = -6000 \\ \dfrac{M_{BC}^{o}}{x} = \dfrac{M_{CB}^{o}}{x} = 0; \qquad \dfrac{M_{ED}^{o'}}{x} = 0 \\ \dfrac{M_{CF}^{o'}}{x} = 3E(2Kr)\left[0-(-\alpha)\right] = 6EKr\alpha = 6000 \\ \dfrac{M_{CD}^{o'}}{x} = 3E(2Kr)\left[0-\dfrac{5}{4}\alpha\right] = -7{,}5EKr\alpha = -7500 \end{cases}$$

5.3.- $M_i^{Fij} = ?$ $\qquad \left\{ \dfrac{M_B^{Fij}}{x} = -6000; \right. \qquad \left\{ \dfrac{M_C^{Fij}}{x} = -1500 \right.$

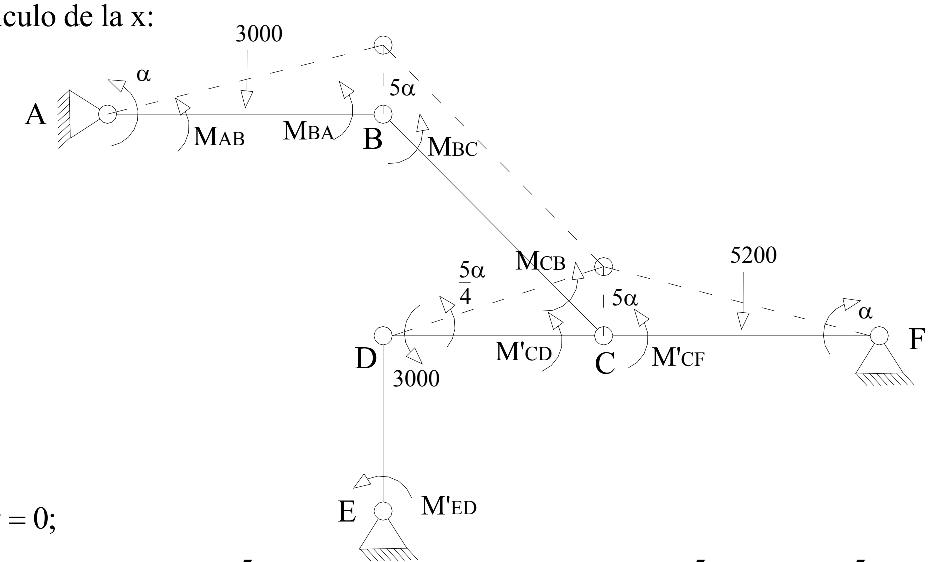
6.-

Junta	A	B		C		
Miembro	AB	BA	BC	CB	CD	CF
D_{ij}	0	0,4	0,6	0,333	0,333	0,333
M_{ij}^{0} (1)	780	-1720	2030	1530	1545	4875
			-1325	-2650	-2650	-2650
	202	405	610	305		
			-51	-102	-101	-102
	10	21	30	15		
			-3	-5	-5	-5
	0	1	2	1	0	0
M_{ij} (1)	992	-1293	1293	-906	-1211	2118
$\frac{1}{x}M_{ij}^{o}$ (2)	-6000	-6000	0	0	-7500	6000
	1200	2400	3600	1800		
			-50	-100	-100	-100
	10	20	30	15		
			-3	-5	-5	-5
	0	1	2	0	0	0
$\frac{1}{x}M_{ij}$ (2)	-4790	-3579	3579	1710	-7605	5895

7.- Superposición:

$$\begin{cases} M_{AB} = 992 - 4790x; \\ M_{BC} = 1293 + 3579x; \\ M'_{CD} = -1211 - 7605x; \\ M'_{ED} = 900 \end{cases} \qquad \begin{cases} M_{BA} = -1293 - 3579x \\ M_{CB} = -906 + 1710x \\ M'_{CF} = 2118 + 5895x \end{cases}$$

8.- Cálculo de la x:



$$\sum Tv = 0;$$

$$(M_{AB} + M_{BA})\alpha - 3000(\frac{5}{2}\alpha) + M'_{CF}(-\alpha) + (M'_{CD} + 3000)(\frac{5}{4}\alpha) - 5200(\frac{5}{2}\alpha) = 0$$

$$(-301 - 8369x) - 7500 - 2118 - 5895x + 2236,25 - 9506,25x - 13000 = 0$$

$$-23770,25x - 20682,75 = 0$$

$$x = -0,87011$$

Momentos Definitivos: $\begin{cases} M_{AB} = 5160; \\ M_{BC} = -1821; \\ M'_{CD} = 5406; \\ M'_{ED} = 900 \end{cases}$ $\begin{cases} M_{BA} = 1821 \\ M_{CB} = -2394 \\ M'_{CF} = -3011 \end{cases}$

Corrección: $\begin{cases} M^C_{AB} = 4380; \\ M^C_{BC} = -3851; \\ M'^C_{CD} = 3861; \end{cases}$ $\begin{matrix} M^C_{BA} = 3541 \\ M^C_{CB} = -3924 \\ M'^C_{CF} = -7886 \end{matrix}$

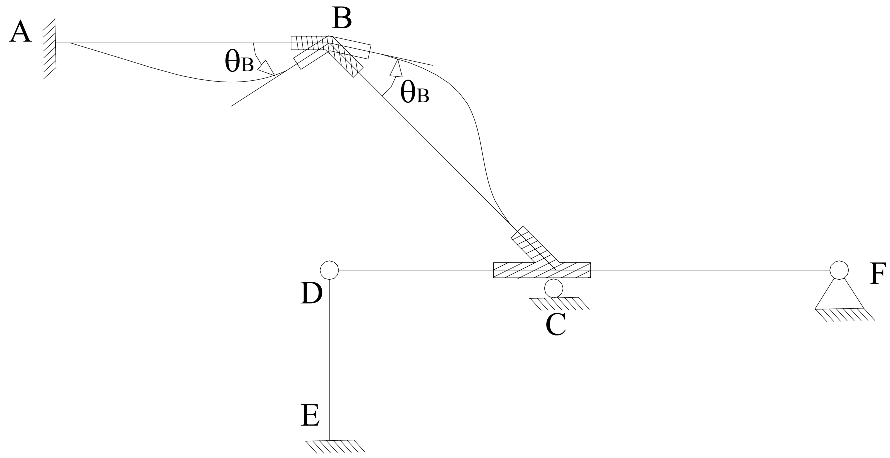
$$\psi_{AB} = \frac{-870}{EKr}; \qquad \psi_{CF} = \frac{870}{EKr}; \qquad \psi_{CD} = \frac{-1087,5}{EKr}$$

$$\theta_{CB} = \frac{-3924 - \frac{1}{2}(-3851)}{3E(\frac{3Kr}{2})} = \frac{-444,11}{EKr}; \qquad \theta_{CF} = \frac{-7886}{3E(2Kr)} + \frac{870}{EKr} = \frac{444,33}{EKr}$$

$$\theta_{CD} = \frac{3861}{3E(2Kr)} - \frac{1087,5}{EKr} = \frac{444}{EKr}$$

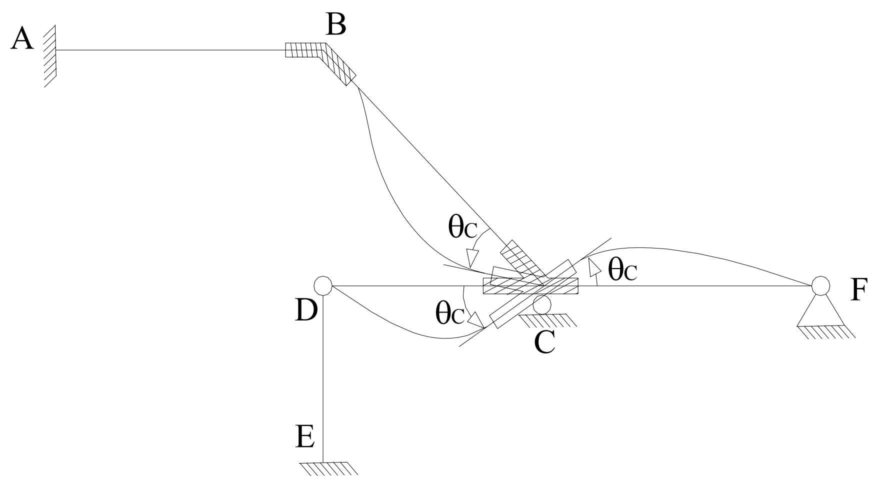
Parte Complementaria para el Método de las Rotaciones.

Problema Complementario # 1: $\exists\theta_B$; $\psi_{ij} = 0$



$$\begin{cases} M_{AB} = 2EKr\theta_B = 2x; \\ M_{BC} = 6EKr\theta_B = 6x; \\ M'_{CD} = M'_{CF} = M'_{ED} = 0 \end{cases} \qquad \begin{cases} M_{BA} = 4EKr\theta_B = 4x \\ M_{CB} = 3EKr\theta_B = 3x \end{cases}$$

Problema Complementario # 2: $\exists\theta_C$; $\psi_{ij} = 0$



$$\begin{cases} M_{AB} = M_{BA} = 0; \\ M_{BC} = 3EKr\theta_C = 3y; \\ M'_{CD} = 6EKr\theta_C = 6y; \end{cases} \qquad \begin{cases} M'_{ED} = 0 \\ M_{CB} = 6EKr\theta_C = 6y \\ M'_{CF} = 6EKr\theta_C = 6y \end{cases}$$

Problema Complementario # 3:

Etapa de desplazabilidad.

$\rightarrow$ Se toman los resultados obtenidos del Método de Cross para el cálculo de los ψ_{ij} en el problema # 2.

$$\left\{\psi_{AB}=\alpha;\right. \qquad \left\{\psi_{DC}=\frac{5}{4}\alpha;\right. \qquad \left\{\psi_{CF}=-\alpha\right.$$

$$\begin{cases} M_{AB}=M_{BA}=-6EKr\alpha=-6z & M'_{ED}=0 \\ M_{BC}=M_{CB}=0 & \\ M'_{CD}=-7,5EKr\alpha=-7,5z; & M'_{CF}=6EKr\alpha=6z \end{cases}$$

Superposición:

$$\begin{cases} M_{AB}=2x-6z+780; & M_{BA}=4x-6z-1720 \\ M_{BC}=6x+3y+1880; & M_{CB}=3x+6y+1680 \\ M'_{CD}=6y-7,5z+1545; & M'_{CF}=6y+6z+4875 \\ M'_{ED}=900 & \end{cases}$$

Ecuaciones de Equilibrio:

Para $\theta_B \rightarrow \sum M_B=0$

$$M_{BA}+M_{BC}-M_B^{Ext}=0; \qquad M_B^{Ext}=0$$
$$10x+3y-6z+160=0$$

Para $\theta_C \rightarrow \sum M_C=0$

$$M_{CB}+M'_{CD}+M'_{CF}-M_C^{Ext}=0; \qquad M_C^{Ext}=0$$
$$3x+18y-1,5z+8100=0$$

Para $v_C \rightarrow$ Se toma la ecuación obtenida para el cálculo de la x aplicando trabajo virtual en el Método de Cross.

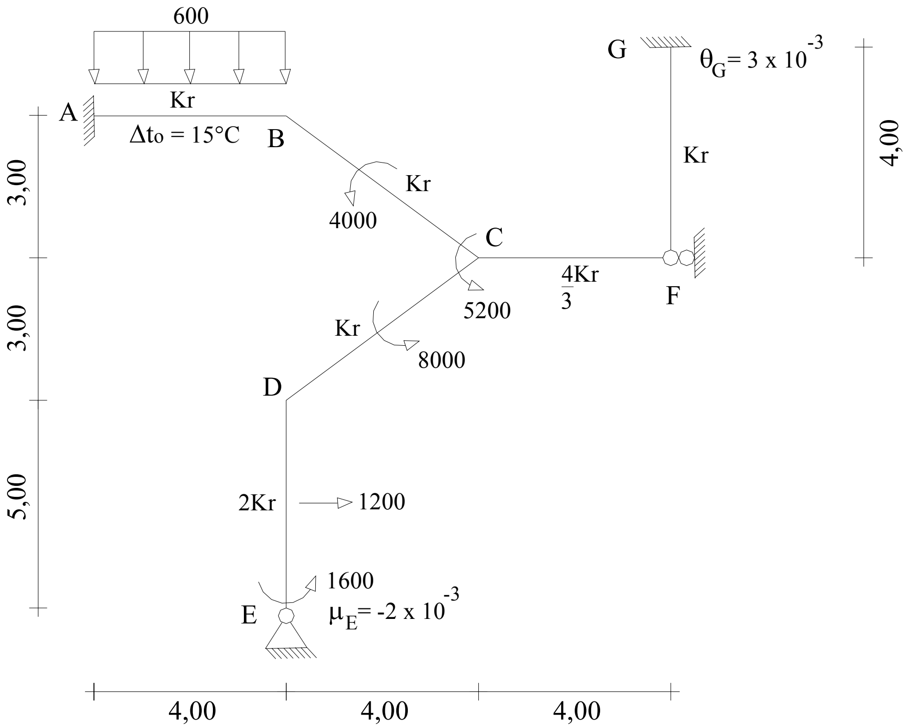
$$\sum Tv=0$$
$$(M_{AB}+M_{BA})-7500-M'_{CF}+(M'_{CD}+3000)(5/4)-13000=0$$
$$6x-12z-940-7500-6y-6z-4875+(6y-7,5z+4545)5/4-13000=0$$
$$6x-6y-18z-26315+7,5y-9,375z+5681,25=0$$
$$6x+1,5y-27,375z-20633,75=0$$

$$\begin{cases} x=-398,9939 \\ y=-455,6813 \\ z=-866,1635 \end{cases}$$

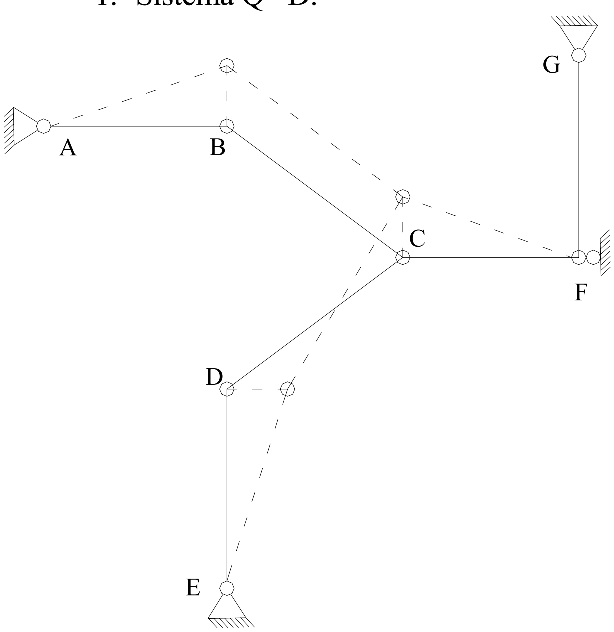
8.- Resolver:

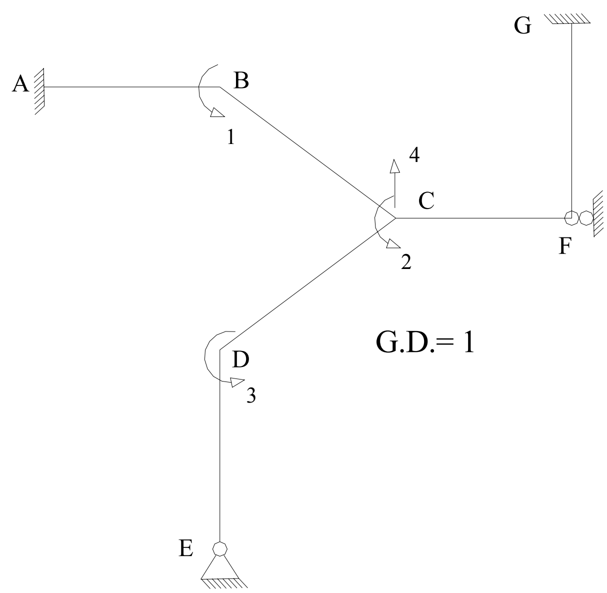
a.- Método de Cross.

b.- Método de las Rotaciones.

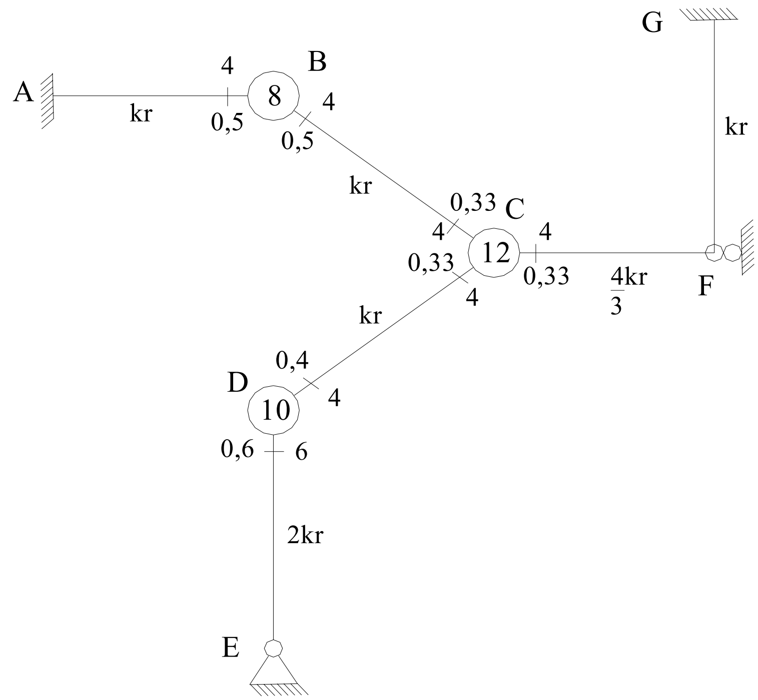


1.- Sistema Q –D.





2.- D_{ij} y r_{ij}



3.- Problema # 1.

4.- Solución del Problema # 1.

4.1.- Problema Primario: $\theta_B = \theta_C = \theta_D = 0$

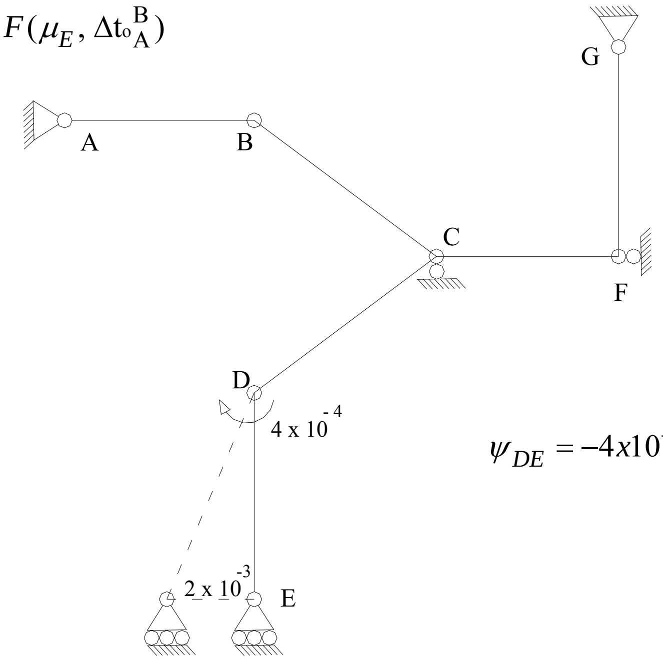
4.1.1.- $M_{ij}^E = F(\text{cargas, } \Delta(\Delta t));\qquad \Delta t = 0$

a.- F (cargas)

$$\begin{cases} M_{AB}^E = -M_{BA}^E = \dfrac{600(4^2)}{12} = 800; \\ M_{ED}^E = -M_{DE}^E = \dfrac{1200(5)}{8} = 750; \\ M_{CF}^E = M_{FC}^E = 0; \end{cases} \qquad \begin{cases} M_{BC}^E = M_{CB}^E = \dfrac{4000}{4} = 1000 \\ M_{CD}^E = M_{DC}^E = \dfrac{8000}{4} = 2000 \\ M_{FG}^E = M_{GF}^E = 0 \end{cases}$$

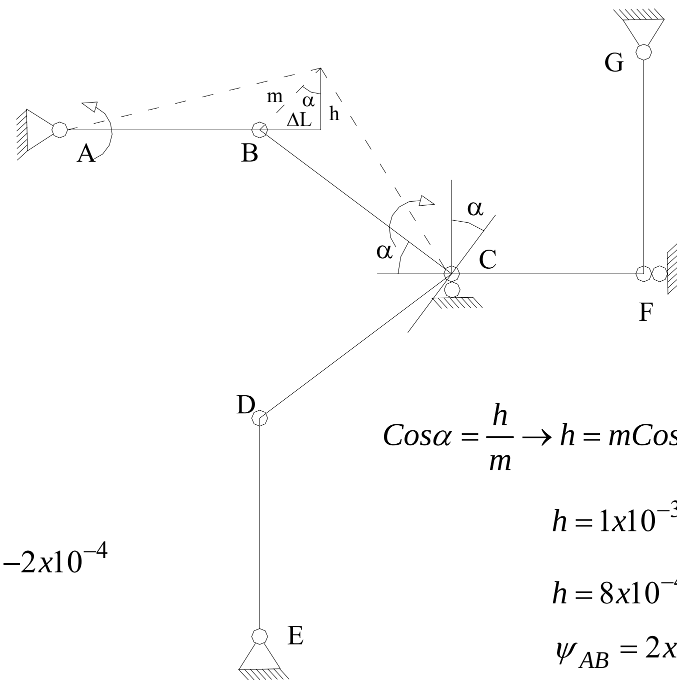
4.1.2.- $\psi_{ij}^{o} = F(\mu_E, \Delta t^{\circ}{}_{A}^{B})$

$F(\mu_E)$



$\psi_{DE} = -4x10^{-4}$

$F(\Delta t^{\circ}{}_{A}^{B})$



$sen\alpha = 3/5$

$\cos\alpha = 4/5$

$(\Delta L = 6x10^{-4})$

$$Sen\alpha = \frac{\Delta L}{m} \rightarrow m = \frac{\Delta L}{Sen\alpha}$$

$$m = \frac{6x10^{-4}}{3/5}$$

$$m = 1x10^{-3}$$

$$\psi_{CB} = \frac{-m}{5} = -2x10^{-4}$$

$$Cos\alpha = \frac{h}{m} \rightarrow h = mCos\alpha$$

$$h = 1x10^{-3}\,\frac{4}{5}$$

$$h = 8x10^{-4}$$

$$\psi_{AB} = 2x10^{-4}$$

Definitivos:
$$\begin{cases} \psi_{AB}^{o} = 2x10^{-4} \\ \psi_{BC}^{o} = -2x10^{-4} \\ \psi_{DE}^{o} = -4x10^{-4} \end{cases}$$

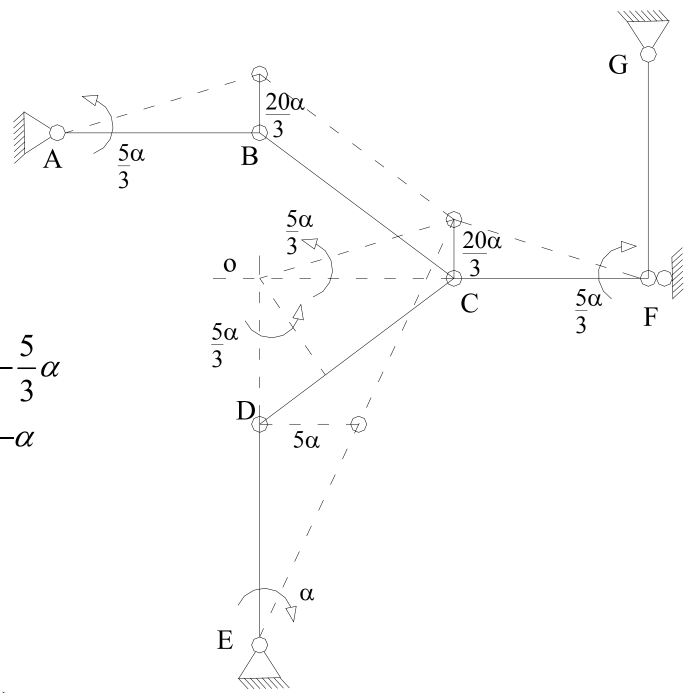
$M_{ij}^{o} = ?$

$$\begin{cases} M_{AB}^{o} = 800 - 6EKr(2x10^{-4}) = 680 \\ M_{BA}^{o} = -800 - 6EKr(2x10^{-4}) = -920 \\ M_{BC}^{o} = 1000 - 6EKr(-2x10^{-4}) = 1120 \\ M_{CB}^{o} = 1000 - 6EKr(-2x10^{-4}) = 1120 \\ M_{CD}^{o'} = 2000; \qquad M_{DC}^{o'} = 2000 \\ M_{DE}^{o'} = -750 + \frac{1}{2}(1600 - 750) + 3E(2Kr)(0 + 4x10^{-4}) = -85 \\ M_{CF}^{o'} = 3EKr(3x10^{-3}) = 900; \qquad M_{GF}^{o'} = 0 \end{cases}$$

$M^{Fij} = ?$ $\quad \left\{ M_{B}^{Fij} = 200; \right. \quad \left\{ M_{C}^{Fij} = -2080; \right. \quad \left\{ M_{D}^{Fij} = 1915 \right.$

5.- Problema # 2.

5.1.- $\psi_{ij}^{o} = F(\text{trasl. Apoyo})$



$$\begin{cases} \psi_{AB} = \frac{5}{3}\alpha; \\ \psi_{DC} = \frac{5}{3}\alpha; \end{cases} \qquad \begin{cases} \psi_{CF} = -\frac{5}{3}\alpha \\ \psi_{DE} = -\alpha \end{cases}$$

5.2.- $M_{ij}^{o} = F(\psi_{ij})$

Sea $(EKr\alpha = 1000x)$

$$\begin{cases} M_{AB}^{o} = M_{BA}^{o} = -6EKr(5/3\alpha) = -10EKr\alpha \\ M_{BC}^{o} = M_{CB}^{o} = 0; \qquad M_{GF}^{o'} = 0 \\ M_{CF}^{o'} = 3E(4/3Kr)(0 + 5/3\alpha) = 20/3EKr\alpha \\ M_{CD}^{o} = M_{DC}^{o} = -6EKr(5/3\alpha) = -10EKr\alpha \\ M_{DE}^{o'} = 3E(2Kr)(0 + \alpha) = 6EKr\alpha \end{cases}$$

$$\begin{cases} M^{o}_{AB}/x = M^{o}_{BA}/x = -10000; \\ M^{o'}_{CF}/x = 20000/3; \\ M^{o'}_{DE}/x = 6000; \end{cases} \qquad \begin{cases} M^{o}_{BC}/x = M^{o}_{CB}/x = 0 \\ M^{o}_{CD}/x = M^{o}_{DC}/x = -10000 \\ M^{o'}_{GF}/x = 0 \end{cases}$$

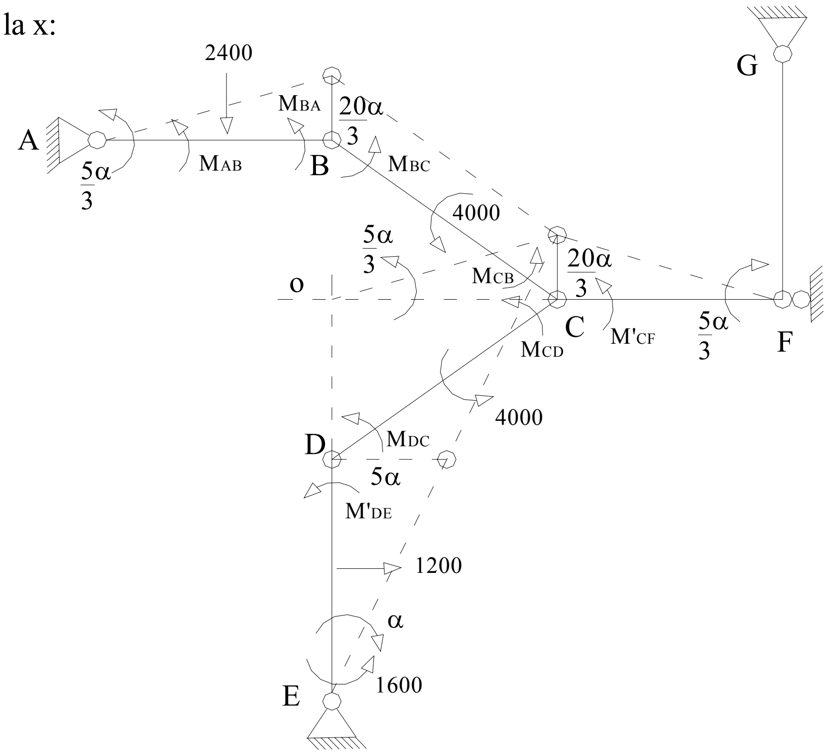
5.3.- $M_i^{Fij} = ?$ $\qquad \left\{ M_B^{Fij}/x = -10000; \quad M_C^{Fij}/x = -10000/3; \quad M_D^{Fij}/x = -4000 \right.$

6.- Etapa Complementaria

Junta	A	B			C		D	
Miembro	AB	BA	BC	CB	CF	CD	DC	DE
D_{ij}	0	0,5	0,5	1/3	1/3	1/3	0,4	0,6
M^{0}_{ij} (1)	680	-920	1120	1120	0	2000	2000	-85
			347	694	692	694	347	
	-137	-273	-274	-137		-452	-904	-1358
			98	196	197	196	98	
M^{C}_{ij} (1)	-25	-49	-49	-24		-20	-40	-58
			8	15	14	15	8	
	-2	-4	-4	-2		-1	-3	-5
			0	1	1	1	0	
M_{ij} (1)	516	-1246	1246	1863	904	2433	1506	-1506
$\frac{1}{x}M^{o}_{ij}$ (2)	-10000	-10000	0	0	20000/3	-10000	-10000	6000
	2500	5000	5000	2500				
			139	278	277	278	139	
	-34	-69	-70	-35		772	1544	2317
			-123	-246	-245	-246	-123	
M^{C}_{ij} (2)	30	61	62	31		25	50	73
			-9	-18	-20	-18	-9	
	3	5	4	2		2	4	5
			0	-1	-2	-1	0	
$\frac{1}{x}M_{ij}$ (2)	-7501	-5003	5003	2511	6680	-9188	-8395	8395

7.- Superposición: $\begin{cases} M_{AB} = 516 - 7501x; \\ M_{BC} = 1246 + 5003x; \\ M'_{CF} = 904 + 6680x; \\ M_{DC} = 1506 - 8395x; \end{cases}$ $\begin{cases} M_{BA} = -1246 - 5003x \\ M_{CB} = 1863 + 2511x \\ M_{CD} = 2433 - 9188x \\ M'_{DE} = -1506 + 8395x \end{cases}$

8.- Cálculo de la x:



$$\sum Tv = 0;$$

$$(M_{AB} + M_{BA} - 2400(2))\frac{5}{3}\alpha + (M_{BC} + M_{CB} + 4000)(0\alpha) + M'_{CF}(\frac{-5}{3}\alpha) +$$

$$+ (M_{CD} + M_{DC} + 8000)\frac{5}{3}\alpha + (M'_{DE})(-\alpha) + 1200(2,5\alpha) - 1600\alpha = 0$$

$$(-5530 - 12504x)\frac{5}{3} - (904 + 6680x)\frac{5}{3} + (11939 - 17583x)\frac{5}{3} + 1506 -$$

$$-8395x + 3000 - 1600 = 0$$

$$(5505 - 36767x)\frac{5}{3} + 2906 - 8395x = 0$$

$$9175 - 61278,333x + 2906 - 8395x = 0$$

$$12081 - 69673,333x = 0$$

$$x = 0,1734$$

Momentos Definitivos:
$$\begin{cases} M_{AB} = -785; \\ M_{BC} = 2114; \\ M'_{CF} = 2062; \\ M_{DC} = 50; \end{cases} \qquad \begin{cases} M_{BA} = -2114 \\ M_{CB} = 2298 \\ M_{CD} = 840 \\ M'_{DE} = -50 \end{cases}$$

Comprobación:
$$\begin{cases} M^C_{AB} = -1465; & M^C_{BA} = -1194 \\ M^C_{BC} = 994; & M^C_{CB} = 1178 \\ M'^C_{CF} = 2062; & M^C_{CD} = -1160 \\ M^C_{DC} = -1950; & M'^C_{DE} = 35 \end{cases}$$

$$EKr\alpha = 1000x \rightarrow \alpha = \frac{1000x}{EKr}$$

$$\psi_{AB} = \frac{289}{EKr}; \qquad \psi_{DC} = \frac{289}{EKr};$$

$$\psi_{CF} = \frac{-289}{EKr}; \qquad \psi_{DE} = \frac{-173,4}{EKr};$$

$$\theta_{CB} = \frac{1178 - \frac{1}{2}(994)}{3E(Kr)} = \frac{227}{EKr}; \qquad \theta_{CF} = \frac{2062}{3E(\frac{4}{3}Kr)} - \frac{289}{EKr} = \frac{226,5}{EKr}$$

$$\theta_{CD} = \frac{-1160 - \frac{1}{2}(-1950)}{3EKr} + \frac{289}{EKr} = \frac{227,33}{EKr}$$

$$\theta_{BA} = \frac{-1194 - \frac{1}{2}(-1465)}{3E(Kr)} + \frac{289}{EKr} = \frac{135,167}{EKr}; \qquad \theta_{BC} = \frac{994 - \frac{1}{2}(1178)}{3E(Kr)} = \frac{135}{EKr}$$

$$\theta_{DC} = \frac{-1950 - \frac{1}{2}(-1160)}{3E(Kr)} + \frac{289}{EKr} = \frac{-167,667}{EKr}; \qquad \theta_{DE} = \frac{35}{3E(2Kr)} - \frac{173,4}{EKr} = \frac{-167,567}{EKr}$$

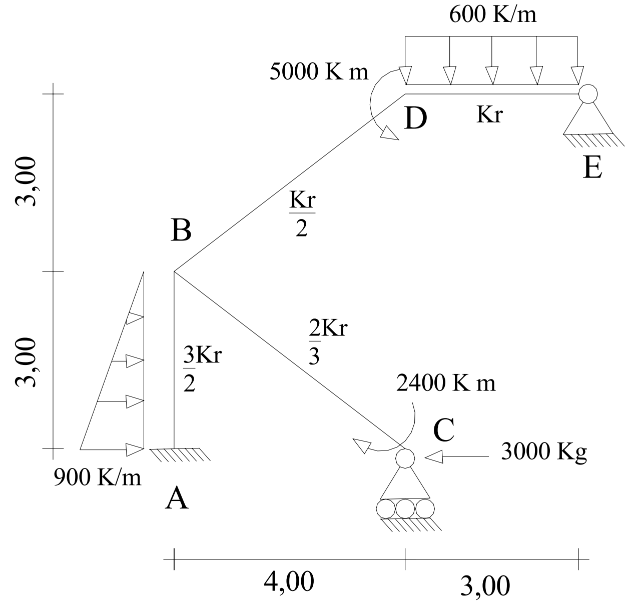
9.- Analizar haciendo uso de:

a.- Método de las Rotaciones.

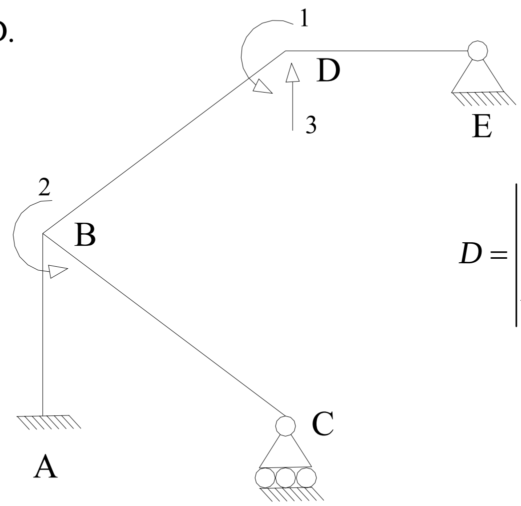
b.- Método de Cross.

$\nu_A = -2x10^{-2}\,m;$ $\Delta t_o^{DE} = 10°C;$

$\alpha t = 1x10^{-5}\ 1/°C;$ $EKr = 10^5\,K.m$



1.- Sistema Q –D.



$$D = \begin{vmatrix} \theta_B \\ \theta_D \\ \nu_D \end{vmatrix}$$

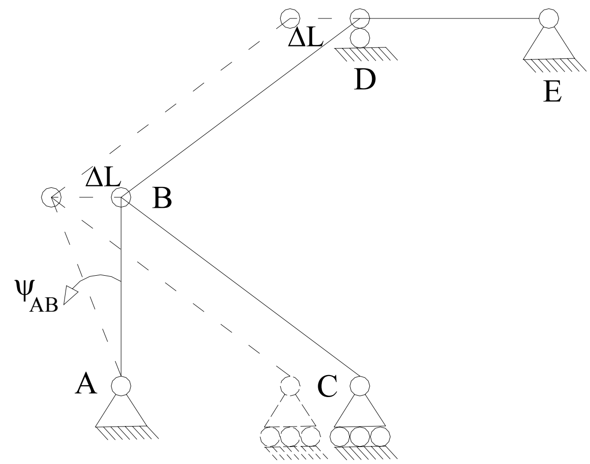
2.- Problema Primario (1).

$$\theta_B = \theta_D = \nu_D = 0$$

2.1.- $M_{ij}^{Comp} = ?$
$$\begin{cases} M_{AB}^{E} = 405; & M_{BA}^{E} = -270 \\ M_{DE}^{E} = 450; & M_{ED}^{E} = -450 \end{cases}$$

2.2.- $\psi_{ij} = F(\nu_A, \Delta t_o^{DE})$

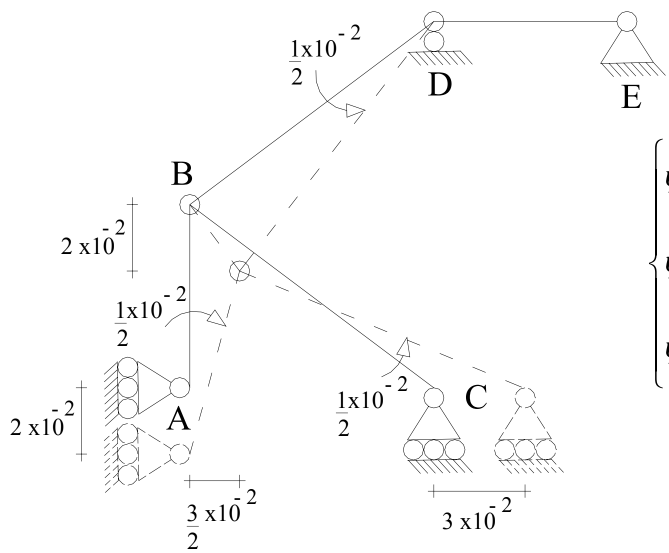
2.2.1.- $\psi_{ij} = F(\Delta t_o)$



$\Delta L = 10^{-5}.10.3 = 3x10^{-4}$

$\psi_{AB} = \dfrac{3x10^{-4}}{3} = 10^{-4}$

2.2.2.- $\psi_{ij} = F(\nu_A)$



$$\begin{cases} \psi_{AB} = \dfrac{-1}{2}x10^{-2}; \\ \psi_{BD} = \dfrac{1}{2}x10^{-2}; \\ \psi_{BC} = \dfrac{1}{2}x10^{-2} \end{cases}$$

2.2.3.- $\psi_{ij}^{Def} = ?$

$$\begin{cases} \psi_{AB} = 1x10^{-4} + (-1/2x10^{-2}) = -4,9x10^{-3} \\ \psi_{BD} = 1/2x10^{-2}; \qquad \psi_{BC} = 1/2x10^{-2} \end{cases}$$

2.3.- $M_{ij}^{o} = ?$

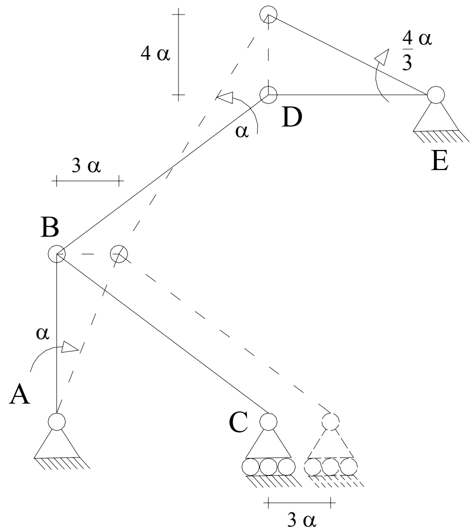
$$\begin{cases} M_{AB}^{o} = 405 - 6E(3/2Kr)(-4,9x10^{-3}) = 4815 \\ M_{BA}^{o} = -270 + 4320 = 4140 \\ M_{BD}^{o} = -6E(Kr/2)1/2x10^{-2} = -1500 \\ M_{DB}^{o} = -1500 \\ M_{BC}^{o'} = 1/2(-2400) - 3E(2/3Kr)1/2x10^{-2} = -2200 \\ M_{DE}^{o'} = 450 + 1/2(450) = 675 \end{cases}$$

2.4.- $M_{i}^{Fij} = ?$ $\quad \{M_{B}^{Fij} = 440; \quad \{M_{D}^{Fij} = -5825$

3.- Problema # 2.

3.1.-Problema Primario.

3.1.1.- $\psi_{ij} = F(\nu_D)$



$$\begin{cases} \psi_{AB} = -\alpha; \\ \psi_{BC} = 0; \\ \psi_{BD} = \alpha; \\ \psi_{DE} = \dfrac{-4}{3}\alpha \end{cases}$$

3.1.2.-

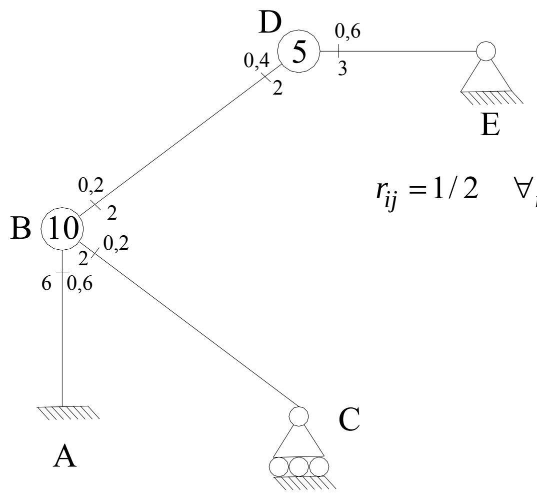
$$\begin{cases} M^o_{AB} = M^o_{BA} = -6E(3/2Kr)(-\alpha) = 9EKr\alpha \\ M^o_{BD} = M^o_{DB} = -6E(Kr/2)\alpha = -3EKr\alpha \\ M^{o'}_{BC} = -3E(2/3Kr)0 = 0 \\ M^{o'}_{DE} = -3E(Kr)(-4/3\alpha) = 4EKr\alpha \end{cases}$$

$Sea \quad EKr\alpha = 1000x$

$$\begin{cases} \dfrac{M^o_{AB}}{x} = \dfrac{M^o_{BA}}{x} = 9000; \qquad \dfrac{M^o_{BD}}{x} = \dfrac{M^o_{DB}}{x} = -3000 \\ \dfrac{M^{o'}_{BC}}{x} = 0; \qquad \dfrac{M^{o'}_{DE}}{x} = 4000 \end{cases}$$

3.1.3.- $M_i^{Fij} = ?$ $\quad \left\{ \dfrac{M_B^{Fij}}{x} = 6000; \right.$ $\quad \left\{ \dfrac{M_D^{Fij}}{x} = 1000 \right.$

4.- Factores de Distribución y Transporte.



$$r_{ij} = 1/2 \quad \forall_{ij} \neq DE$$
$$\neq BC$$

Junta	A		B			D
Miembro	AB	BA	BC	BD	DB	DE
D_{ij}	0	0,60	0,20	0,20	0,40	0,60
M^0_{ij} (1)	4815	4140	-2200	-1500	-1500	675
				1165	2330	3495
	-481	-963	-321	-321	-160	
				32	64	96
	-10	-20	-6	-6	-3	
				0	1	2
M^0_{ij} (1)	4324	3157	-2527	-630	732	4268
M^o_{ij} (2)	9000	9000	0	-3000	-3000	4000
	-1800	-3600	-1200	-1200	-600	
				-80	-160	-240
	24	48	16	16	8	
				-1	-3	-5
	0	1	0	0	0	
M_{ij} (2)	7224	5449	-1184	-4265	-3755	3755

5.- Aplicando Superposición:

$$\begin{cases} M_{AB} = 4324 + 7224x; \\ M_{BA} = 3157 + 5449x; \\ M_{BD} = -630 - 4265x; \end{cases} \qquad \begin{cases} M_{DB} = 732 - 3755x \\ M_{BC} = -2527 - 1184x \\ M_{DE} = 4268 + 3755x \end{cases}$$

6.- Cálculo de la X:

$$\sum Tv = 0;$$

$$-(M_{AB} + M_{BA})\alpha + (M_{BD} + M_{DB})\alpha - M'_{DE}(4/3\alpha) - 3000(3\alpha) - 1800(2\alpha) + 1350\alpha = 0$$

$$-(4324 + 3157 + 7224x + 5449x) + (732 - 630 - 4265x - 3755x) - (4/3)(4268 + 3755x) -$$
$$-9000\alpha - 3600\alpha + 1350\alpha = 0$$

$$-25700x - 24320 = 0$$

$$x = -0,9463$$

7.- Momentos Definitivos: $\begin{cases} M_{AB} = -2512; \\ M_{BD} = 3406; \\ M_{BC} = -1407; \end{cases} \qquad \begin{cases} M_{BA} = -1999 \\ M_{DB} = 4285 \\ M_{DE} = 715 \end{cases}$

Solución a través del Método de las Rotaciones.

Tenemos:

1.- $\begin{cases} M^{o}_{AB} = 4815; \qquad M^{o}_{BA} = 4140 \\ M^{o}_{BD} = -1500; \qquad M^{o}_{DB} = -1500 \\ M^{o'}_{BC} = -2200; \qquad M^{o'}_{DE} = 675 \end{cases}$

2.- Problema Complementario # 1: $\exists \quad \theta_B$

Sea $(EKr\theta_B = x)$

$$\begin{cases} M_{AB} = 2E(\frac{3}{2}Kr)\theta_B = 3EKr\theta_B = 3x; \\ M_{BA} = 4E(\frac{3}{2}Kr)\theta_B = 6EKr\theta_B = 6x; \\ M_{BD} = 4E(\frac{Kr}{2})\theta_B = 2EKr\theta_B = 2x; \end{cases} \qquad \begin{cases} M_{DB} = 2E(\frac{Kr}{2})\theta_B = EKr\theta_B = x \\ M'_{BC} = 3E(\frac{2}{3}Kr)\theta_B = 2EKr\theta_B = 2x \\ M^{o'}_{DE} = 0 \end{cases}$$

3.- Problema Complementario # 2: $\exists \quad \theta_D$

$$\begin{cases} M_{AB} = M_{BA} = M'_{BC} = 0; \\ M_{BD} = 2E(\frac{Kr}{2})\theta_D = EKr\theta_D = y; \end{cases} \qquad \begin{cases} M_{DB} = 4E(\frac{Kr}{2})\theta_D = 2EKr\theta_D = 2y \\ M'_{DE} = 3EKr\theta_D = 3y \end{cases}$$

4.- Problema Complementario # 3:

Del problema anterior

$$\begin{cases} M_{AB} = M_{BA} = 9EKr\alpha = 9z \\ M_{BD} = M_{DB} = -3EKr\alpha = -3z \\ M'_{BC} = 0; \qquad M'_{DE} = 4EKr\alpha = 4z \end{cases}$$

5.- Aplicando superposición:

$$\begin{cases} M_{AB} = 4815 + 3x + 9z; \\ M_{BA} = 4140 + 6x + 9z; \\ M_{BD} = -1500 + 2x + y - 3z; \end{cases} \qquad \begin{cases} M_{DB} = -1500 + x + 2y - 3z \\ M'_{BC} = -2200 + 2x \\ M'_{DE} = 675 + 3y + 4z \end{cases}$$

6.- Ecuaciones de Equilibrio:

6.1.- $M_{BA} + M_{BD} + M'_{BC} = 0$

$10x + y + 6z = -440 \qquad (1)$

6.2.- $M_{DB} + M'_{DE} - 5000 = 0$

$x + 5y + z = 5825 \qquad (2)$

6.3.-Del problema anterior tenemos:

$$-(M_{AB} + M_{BA})\alpha + (M_{BD} + M_{DB})\alpha - M'_{DE}(\frac{4}{3}\alpha) = 11250\alpha$$

Sustituyendo:

$$-(4815 + 3x + 9z + 4140 + 6x + 9z) + (-3000 + 3x + 3y - 6z) - \frac{4}{3}(675 + 3y + 4z) = 11250$$

$$6x + y + 29,33z = -24105 \qquad (3)$$

$$\begin{bmatrix} 10 & 1 & 6 \\ 1 & 5 & 1 \\ 6 & 1 & 29,33 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -440 \\ 5825 \\ -24105 \end{bmatrix}$$

7.- Resolviendo el Sistema de Ecuaciones:

$$x = 396,34; \qquad y = 1275,01; \qquad z = -946,41$$

8.-Sustituyendo, Momentos Definitivos:

$$\begin{cases} M_{AB} = -2512; & M_{BA} = -1999 \\ M_{BD} = 3406; & M_{DB} = 4285 \\ M_{BC} = -1407; & M_{DE} = 715 \end{cases}$$